

Exponential-Range Mass Production of Boolean Functions

Local recovery and optimal quantum state preparation

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Abstract

The hardest Boolean functions on n bits require $\Theta(2^n/n)$ gates to evaluate once. Must evaluating the same function on t unrelated inputs cost t times as much? We show that every Boolean function can be evaluated on any $t \leq 2^{\gamma n}$ inputs using a total of $O_\gamma(2^n/n)$ gates, for every fixed $0 \leq \gamma < 1$. Thus exponentially many evaluations fit within a constant factor of the worst-case cost of one. In the standard AND, OR, NOT basis, our bound is $(1/(1-\gamma) + o_\gamma(1))2^n/n$ gates. Uhlig previously achieved $(1 + o(1))2^n/n$ gates for the smaller range $t = 2^{o(n/\log n)}$. We also attain $O_\gamma(2^n/n)$ gates and $O_\gamma(n)$ depth simultaneously.

Our construction encodes shorter restrictions of the function and recovers the requested outputs from disjoint sets of encoded functions. The gate bound includes choosing those sets, evaluating the functions, and routing the answers. Given the truth table, a randomized algorithm constructs circuits attaining the simultaneous size and depth bounds in polynomial time in the table length, with high probability. Efficient deterministic construction of their fixed scheduling menus remains open. A Lean companion verifies the recursive and sharp-coefficient Boolean size bounds.

As an application, we determine the optimal worst-case cost of preparing t copies of a specified n -qubit pure state throughout the same range: $\Theta_\gamma((2^n/n) \log_2(t+1))$ elementary Clifford+ T gates. The error is at most $1/10$ in trace distance for the entire batch, and scratch qubits are restored to zero. Coherent Boolean evaluation gives the upper bound; a packing and circuit-counting argument gives the matching lower bound.

1 Introduction

How much circuit size can be saved when the same function must be evaluated on many unrelated inputs? Each output depends on a separate input block, but the circuit may reorganize and share intermediate computations across copies. This is the direct-sum question studied here; the input values may repeat.

For a Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ and an integer $t \geq 1$, define

$$f^{\times t}(x^{(1)}, \dots, x^{(t)}) = (f(x^{(1)}), \dots, f(x^{(t)})).$$

The naive construction uses t disjoint copies of a circuit for f and gives

$$C(f^{\times t}) \leq tC(f).$$

Here C counts elementary Boolean gates. The classical Shannon–Lupanov bounds say that the hardest functions on n bits have circuit size $\Theta(2^n/n)$; in our basis the worst-case size is $(1+o(1))2^n/n$ [Sha49, Lup58, FM05]. We call this the worst-case cost of one evaluation. The mass-production problem asks how large t can become before the cost must rise above that scale. For unrestricted circuits, the naive direct-sum bound can be far from optimal. Uhlig’s theorem allows $t = 2^{o(n/\log n)}$ independent evaluations with no asymptotic increase in the leading worst-case one-copy synthesis cost [Uhl92, Weg87, Kre23].

We prove, by a different construction, that the order-of-growth plateau reaches $2^{\gamma n}$ simultaneous evaluations for every fixed $\gamma < 1$. The asymptotic coefficient is at most $1/(1-\gamma)$ in our basis. This does not preserve Uhlig’s coefficient 1 for positive fixed γ .

1.1 Boolean size and depth bounds

We fix the standard basis $\{\wedge, \vee, \neg\}$, with fan-in two for the binary gates. Constant sources, fan-out, and output designations are free; circuit size counts the $\wedge$, $\vee$, and $\neg$ gates. Any other fixed finite complete basis changes the result by at most a constant-factor simulation.

Theorem 1.1 (Exponential-range mass production). *For every fixed $0 \leq \gamma < 1$, there is a constant A_γ such that, for every integer $n \geq 1$, every Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$, and every integer $1 \leq t \leq 2^{\gamma n}$,*

$$C(f^{\times t}) \leq A_\gamma \frac{2^n}{n}.$$

The next theorem refines the leading constant in this bound.

Theorem 1.2 (Quantitative leading coefficient). *For every fixed $0 \leq \gamma < 1$, uniformly over Boolean functions on n bits and integers $1 \leq t \leq 2^{\gamma n}$,*

$$C(f^{\times t}) \leq \left(\frac{1}{1-\gamma} + o_\gamma(1) \right) \frac{2^n}{n}.$$

Equivalently, for every $\varepsilon > 0$, the coefficient $1/(1-\gamma) + \varepsilon$ suffices for all sufficiently large n .

The order-of-growth size bound can also be attained at linear depth.

Theorem 1.3 (Simultaneous size and depth). *For every fixed $0 \leq \gamma < 1$, there are constants A_γ, B_γ such that, for every $n \geq 1$, every Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$, and every integer $1 \leq t \leq 2^{\gamma n}$, there is a circuit C computing $f^{\times t}$ with*

$$\text{size}(C) \leq A_\gamma \frac{2^n}{n}, \quad \text{depth}(C) \leq B_\gamma n.$$

This simultaneous bound preserves the order of size in Theorem 1.1; its proof does not retain the leading coefficient of Theorem 1.2. Corollary 10.4 gives a randomized algorithm that, given the full truth table, constructs such a circuit in time polynomial in 2^n , with probability at least $1 - 2^{-n}$ of correctness on every input batch.

For fixed γ , the bound is independent of the batch size throughout the stated range. At the largest allowed batch size, the amortized size per requested value is therefore

$$\frac{C(f^{\times t})}{t} = O_\gamma \left(\frac{2^{(1-\gamma)n}}{n} \right).$$

This is a worst-case synthesis guarantee. For a particular function with a small circuit, the replication bound $tC(f)$ may be smaller; one may always take the better of the two bounds.

There must nevertheless be a transition before the 2^n -copy scale. If $n \geq 2$ and f depends on all n variables, every circuit for $f^{\times t}$ has size at least $t(n-1)$. Hence a uniform $O(2^n/n)$ plateau cannot persist once $t = \omega(2^n/n^2)$; see [Section 12](#). Our theorem reaches every fixed exponential rate, but leaves this near-endpoint regime open.

1.2 An optimal quantum state preparation tradeoff

Mass production also matters when many copies of the same quantum state are needed. A synthesis circuit may be designed for the specified state, and sharing can occur while the copies are prepared from all-zero inputs. Kretschmer established quantum analogues of Uhlig’s theorem by mass producing conditional rotations and using them to prepare states [\[Kre23\]](#). Building on this framework and the connection to coherent data loading [\[HKW25\]](#), our stronger Boolean theorem yields an upper bound for approximate state preparation in a finite gate set throughout the exponential copy range. A packing lower bound gives the matching tradeoff.

Let $M(n, t)$ be the maximum, over n -qubit pure states $|\psi\rangle$, of the minimum number of elementary Clifford+ T gates needed to prepare $|\psi\rangle^{\otimes t}$ to trace-distance error at most $1/10$. The error is measured on all nt output qubits together: no measurement on the batch can change an outcome probability by more than $1/10$. Gates may connect any qubits; scratch qubits start and end in zero. We count every gate, including Clifford gates, and permit circuits tailored to ψ . [Section 11](#) gives the precise model and the proof.

Theorem 1.4 (Optimal approximate quantum state mass production). *For every fixed $0 \leq \gamma < 1$, uniformly over integers $1 \leq t \leq 2^{\gamma n}$ as $n \rightarrow \infty$,*

$$M(n, t) = \Theta_{\gamma} \left(\frac{2^n}{n} \log_2(t+1) \right).$$

The cost grows logarithmically with the number of copies. In particular, it is $\Theta(2^n/n)$ for one copy, $\Theta(2^n \log n/n)$ for $t = n^a$ with fixed $a > 0$, and $\Theta_{\gamma}(2^n)$ for $t = \lfloor 2^{\gamma n} \rfloor$ with fixed $0 < \gamma < 1$. The single-copy endpoint is already known [\[GKW26\]](#). Our upper bound uses coherent Boolean mass production and carefully allocated rotation precision. The lower bound reflects the fact that t copies distinguish states whose single-copy distance is on the order of $1/\sqrt{t}$.

1.3 How the construction works

Split an input as $x = (a, z)$, with a p -bit prefix a and a d -bit suffix z , where $p + d = n$. The prefix selects one of the 2^p shorter functions $f_a(z) = f(a, z)$. The desired saving would come from using only one circuit for each shorter function. But two inputs with the same prefix may require that function at two different suffixes, and one circuit has only one set of input wires.

Coding supplies alternative ways to answer each request. We replace the restrictions f_a by a larger bank of shorter functions, called *resources*. Each desired value can be recovered from many different sets of resources. A scheduler chooses one set for each request so that no resource is needed twice. Each resource circuit can then receive the one suffix assigned to it, and all requested answers can be reconstructed.

Uhlig serves two requests using opposite ends of an XOR chain, then scales through recursion [Uhl92, Weg87]. We replace that chain with a code offering many recovery choices, and schedule disjoint choices for an entire batch. The key is making the scheduling cheap enough to preserve the savings. Section 4 develops this comparison through a worked example.

The construction has four components.

1. **Pack restrictions.** For every fixed suffix z , encode the values $f(a, z)$ over all prefixes a in a code over $\mathbb{F}_q$. As z varies, each bit of each code coordinate defines a resource function on d variables.
2. **Recover locally.** The code coordinates are points in $\mathbb{F}_q^\ell$. A requested symbol is the sum of the $q - 1$ other symbols on any line through its information point. About $q^{\ell-1}$ directions are available, giving many recovery options.
3. **Schedule disjoint recovery.** Test a fixed menu of direction assignments and choose one that serves at least half the pending requests without collisions. Reserve those recovery sets and repeat on the remaining requests. The total gate cost is $\tilde{O}_\ell(tq)$, including selection of a schedule for the live inputs.
4. **Evaluate the resources.** Each code coordinate is now queried on at most one suffix. Apply ordinary one-copy synthesis to each resource function on $d = |z|$ variables, route its value to the requesting line, and decode.

The encoding defines the resource truth tables when the circuit is built. At evaluation time, the circuit selects recovery sets, routes suffixes, evaluates the fixed resource circuits, and decodes. The menus are fixed in advance and work for every batch; the input prefixes determine which entries the circuit selects. The depth refinement in Section 10 increases the progress per phase so that only constantly many phases remain.

1.4 Where the leading coefficient comes from

The main cost is the product of the number of resource functions and the cost of synthesizing one of them. A code whose rate tends to one, together with packing that leaves only a vanishing fraction of its capacity unused, gives

$$\underbrace{(1 + o(1))2^p}_{\text{Boolean resources}} \cdot \underbrace{(1 + o(1))\frac{2^d}{d}}_{\text{gates per resource}} = (1 + o(1))\frac{2^n}{d}.$$

Leaving more suffix bits makes this product smaller, but leaves fewer prefix bits to provide distinct recovery options. The scheduler can accommodate $t \leq 2^{\gamma n}$ while leaving $d = \delta n + O(1)$ suffix bits for any fixed $0 < \delta < 1 - \gamma$, by taking the geometric dimension ℓ sufficiently large and then fixed. Its scheduling and routing costs are exponentially smaller than $2^n/n$. The displayed product therefore gives the coefficient $1/\delta$. Choosing δ close enough to $1 - \gamma$ gives $1/(1 - \gamma) + \varepsilon$ for any fixed $\varepsilon > 0$. For example, the bound for $t \leq 2^{n/2}$ has coefficient $2 + o(1)$. Appendix D explains when this coefficient is forced by the resource-bank architecture.

Machine-checked companion. A Lean 4 companion verifies the recursive size bound and the complete nonuniform construction yielding Theorem 1.2, including its real-rate and additive-error

quantifiers. [Appendix C](#) records the theorem endpoints, pinned sources, and finite accounting. The depth, randomized-construction, quantum, and lower-bound results are written proofs outside this companion.

Development methodology. Generative-AI systems materially assisted mathematical experiments, literature discovery, and proof development, as detailed in the AI Disclosure at the end of the paper.

Organization. [Sections 2](#) and [3](#) compare prior work and fix the circuit model. [Section 4](#) illustrates disjoint recovery with two requests. [Sections 5](#) to [7](#) develop the code, scheduler, and composition bound for the direct proof in [Section 8](#). [Sections 9](#) and [10](#) refine the coefficient and depth; the latter also gives the randomized construction. [Section 11](#) proves the quantum state preparation tradeoff, and [Section 12](#) treats secure evaluation, lower bounds, and open problems.

The appendices give an explicit recursive alternative ([Appendix A](#)), circuit implementation details ([Appendix B](#)), exact accounting for the Lean companion ([Appendix C](#)), and the lower bound for modules with polynomial interfaces ([Appendix D](#)).

2 Relation to earlier work

The table compares the circuit bounds and the main coding antecedents. Here “one-copy leading term” means $(1 + o(1))2^n/n$ gates in our basis, the worst-case cost of evaluating one function once. Uhlig’s 1974 paper [[Uhl74](#)] is a precursor to the mass-production theorem presented in [[Uhl92](#)] and in Wegener’s Theorem 10.2.3 [[Weg87](#)].

Result	Copy range	Circuit-size bound	Comparison
Uhlig [Uhl92 , Weg87 , Kre23]	$\log t = o(n/\log n)$	one-copy leading term	optimal order and leading constant in a smaller copy range
Paul, Lemma 2 [Pau76]	arbitrary t	$O(\max\{n2^n, nt \log^2 t\})$	$O(n2^n)$ throughout every fixed exponential range below one
Holmgren–Rothblum [HR24]	arbitrary t	$O(2^n + tn^3)$ after hardwiring	$O(2^n)$ throughout every fixed exponential range below one
Albrecht, Theorem 1 [Alb89]	$t \leq 2^n$	$O(2^n n^2)$	the full exponential range, with polynomial overhead
Finite-geometry batch codes [IKOS04 , PV19]	code-dependent batch size	disjoint local recovery; no bound on $C(f^{\times t})$	coding antecedents for affine-geometric recovery
This paper	$t \leq 2^{\gamma n}$ for fixed $\gamma < 1$	$O_\gamma(2^n/n)$	optimal order; limiting coefficient at most $1/(1 - \gamma)$

Paul’s construction treats the truth table as an ordered database: sort the input addresses, sweep through the hardwired table, and restore the answers to their original order. For fixed $\gamma < 1$ and

$t \leq 2^\gamma$, $nt \log^2 t = o(n2^n)$, giving the table’s $O(n2^n)$ bound. Paul’s g^k is the same product task considered here; his arbitrary binary Boolean gates can be simulated in our basis at constant cost.

Holmgren and Rothblum prove that Q indexed bits can be selected from a live N -bit database by a bounded-fan-in Boolean circuit of size $O(N + Q \log^3 N)$ and depth $O(\log(N + Q))$ [HR24, Main theorem]. In our nonuniform model, hardwiring the database to the 2^n -bit truth table of f and using the t live n -bit input blocks as indices gives $O(2^n + tn^3)$, as recorded in the table. Thus, throughout every fixed exponential range below one, Theorem 1.1 improves the bounds from Paul’s construction and this multiselection reduction by factors of n^2 and n , respectively. Multiselection also implements the gather step of our routing; Appendix B.3 gives the reduction.

Albrecht studies the same simultaneous-realization problem over a complete binary basis [Alb89, Theorem 1, pp. 53–54]. His construction covers the endpoint $t = 2^n$, with depth $O(n \log n)$ and size a factor $O(n^3)$ above the one-copy scale, as Albrecht also notes.

Hiltgen and Paterson use “mass production” for PI_k -set circuits underlying monotone slice functions [HP95]. Their result gives sharing in that restricted set-circuit setting and leads to an optimal circuit for the Nečiporuk slice. The task here is simultaneous evaluation of an arbitrary Boolean function on disjoint input blocks.

Direct sums in oracle models. Suruga [Sur24, Theorem 1] characterizes limiting amortized worst-case oracle complexity by optimal expected one-instance oracle complexity, for a fixed problem and positive per-instance error. His framework charges oracle calls and leaves internal computation uncharged. Our circuits compute every output exactly, and their savings concern the gates used for that internal computation, including schedule selection and routing.

Coding ingredients and circuit costs. Systematic polynomial evaluation, affine-line recovery, and disjoint recovery sets are established batch-code tools. Section 3.4 of Ishai–Kushilevitz–Ostrovsky–Sahai [IKOS04] chooses random affine lines and deletes intersections while preserving enough points for interpolation. Their Remark 3.11 also observes that the randomized decoding algorithms in that and the following sections can be derandomized using limited independence. In their terminology, the repeated-request analogue relevant here is a primitive multiset batch code. Our local gadget uses the elementary tensor-product evaluation subspace, with a separate degree bound in each variable, whereas their construction uses total-degree Reed–Muller evaluation.

Finite-geometry constructions of primitive batch codes likewise give explicit families of mutually disjoint recovery sets [PV19]. Later lifted-polynomial constructions also use line-based recovery to obtain batch-code guarantees [HPPV20]. Private-information-retrieval codes of Fazeli, Vardy, and Yaakobi form a related family in which each information symbol has many disjoint recovery options [FVY15].

For circuit mass production, selecting recovery sets is itself part of the computation: the prefixes are input wires, and every batch must produce valid recovery records within the gate budget. A generic polynomial bound in the exponentially large code length and batch size need not fit below $2^n/n$. The menu scheduler provides the more precise bound $\tilde{O}_\ell(tq)$, which Proposition 7.1 combines with resource synthesis and routing. Counting occupied-line descriptions keeps the menus short; sharing the occupied-point list across candidates keeps their verification small. Together with high-rate packing, this gives the stated exponential-range bound and its leading coefficient.

Quantum mass production and synthesis. Kretschmer [Kre23, Sections 3–4] gives exact synthesis of $t = 2^{o(n/\log n)}$ copies of any n -qubit state with $O(2^n)$ CNOT gates, allowing arbitrary one-qubit gates. He also obtains an $O(4^n)$ bound for t uses of an arbitrary n -qubit unitary. His reduction from mass production of conditional rotations to state preparation provides the framework for our upper bound. His discussion asks what happens for larger copy ranges. [Theorem 1.4](#) determines the total elementary gate count for $t \leq 2^\gamma$, for every fixed $\gamma < 1$, in a finite gate set at constant error for the whole batch. The larger copy range and matching finite-gate bounds are the advance over his result; the model differences remain essential to the comparison with exact synthesis.

Huggins, Khattar, and Wiebe [HKW25] combine mass production with quantum read-only memory, which loads fixed classical data at addresses held in quantum registers. They give applications to quantum algorithms and analyze both a small-copy plateau and a larger-copy regime. Their Appendix B.5 explains the passage from data loading to state preparation. Our upper proof uses this established connection with the stronger Boolean synthesis bound.

Gosset, Kothari, and Wu [GKW26] determine the optimal T -gate count for state preparation and diagonal-unitary synthesis. Their discussion also records an $O(2^n/n)$ total-gate bound for preparing one state at constant error, and their Appendix B allocates approximation error geometrically across conditional rotations. We count all elementary gates and determine their dependence on t through an exponential copy range. Our proof uses standard conditional-rotation synthesis [SBM06] and finite-gate approximation [DN06], followed by a matching packing lower bound.

3 Circuit conventions and worst-case notation

Our circuits are finite directed acyclic graphs. Source vertices are input bits or the constants zero and one. Internal vertices are labelled by $\wedge$, $\vee$, or $\neg$, with the usual fan-in. A circuit for a map $F : \{0, 1\}^N \rightarrow \{0, 1\}^m$ has m designated output wires, which may be input wires or gate outputs and may repeat. Fan-out, constants, and output designations carry no cost; size is the number of internal gates. We write $C(F)$ for the minimum size of such a circuit. All leading-constant comparisons in this paper refer to this fixed model. The depth of a circuit is the maximum number of internal gates on a path from a source to a designated output, counting a $\neg$ gate as one gate on such a path.

Let

$$L(n) = \max_{f: \{0,1\}^n \rightarrow \{0,1\}} C(f).$$

The Shannon counting lower bound and Lupanov’s synthesis upper bound imply

$$L(n) = \Theta(2^n/n)$$

for every fixed finite complete basis; see [Sha49, Lup58, FM05, Weg87].

Fixing all but one input block and retaining its output gives $C(f^{\times t}) \geq C(f)$. Thus $2^n/n$ is the optimal worst-case order even when several outputs are requested.

Circuits may depend on the full truth table of f . Their size and depth measure evaluation after the circuit has been constructed. Field representations, irreducible polynomials, sorting networks, and scheduling menus may all be fixed as part of that construction. [Corollary 10.4](#) separately bounds

the time to construct the circuits for the simultaneous size and depth theorem from a supplied truth table.

The parameters of the Boolean construction have the following roles. Code length and dimension are measured in field symbols; multiplying by b converts either count to bits.

Parameter	Meaning
$n = p + d, t$	Input length, prefix/suffix split, and number of requests
$q = 2^b, \ell$	Field size, bits per symbol, and fixed geometric dimension
$N = q^\ell, K$	Code length and dimension; K/N is the code rate
$D_q = (q^\ell - 1)/(q - 1)$	Number of line directions through each target
g	Number of requests in the scheduler lemmas
M	Number of slots in a routing table

We use $\tilde{O}_\ell(X)$ for

$$X \cdot \text{poly}_\ell(n, \log q, \log(t + 2), \log(M + 2)).$$

Only polynomial factors are suppressed; in the parameter regimes used below, they never affect an exponential rate.

For a positive integer k , write $[k]_0 = \{0, \dots, k - 1\}$.

We repeatedly use two standard circuit primitives. First, after representing $\mathbb{F}_{2^b}$ in a polynomial basis, field addition, multiplication, inversion of a nonzero element, and comparison of canonical b -bit representations have circuits of size $\text{poly}(b)$. For example, inversion can be implemented by binary exponentiation to the power $2^b - 2$. Second, R records of width w can be sorted by a fixed sorting network using $O(R \log^2 R)$ compare-exchange units, each of size $O(w)$; see [Bat68]. Sorting by a leading type bit, with original positions used as tie breakers when needed, also gives stable fixed-wire compaction within the same asymptotic bound.

4 The two-copy construction as disjoint recovery

Uhlig's two-copy argument illustrates the invariant behind disjoint recovery: each requested value has two representations, and the requests can choose representations that use separate resources.

Fix a prefix length p , put $M = 2^p$, and define the restrictions

$$f_i(z) = f(i, z), \quad i \in \{0, \dots, M - 1\}.$$

Uhlig's two-copy construction introduces the following functions; see the proof of Wegener's Theorem 10.2.3 [Weg87] and the overview in [Kre23, Section 1.2]:

$$g_0 = f_0, \quad g_i = f_{i-1} \oplus f_i \quad (1 \leq i < M), \quad g_M = f_{M-1}.$$

Every restriction has two representations:

$$f_i = \bigoplus_{j=0}^i g_j = \bigoplus_{j=i+1}^M g_j.$$

Given two inputs $(i, z^{(1)})$ and $(j, z^{(2)})$ with $i \leq j$, recover $f_i(z^{(1)})$ from the prefix set $\{0, \dots, i\}$ and $f_j(z^{(2)})$ from the suffix set $\{j+1, \dots, M\}$. The sets are disjoint, so each resource function g_j is evaluated on at most one suffix.

For example, take $p = 2$, so there are four restrictions and five resources. The requests with prefixes 1 and 2 can be answered as

$$f_1(z^{(1)}) = g_0(z^{(1)}) \oplus g_1(z^{(1)}), \quad f_2(z^{(2)}) = g_3(z^{(2)}) \oplus g_4(z^{(2)}).$$

Resources g_0, g_1 receive $z^{(1)}$, resources g_3, g_4 receive $z^{(2)}$, and g_2 is unused. The suffixes can be completely unrelated. Even if the two prefixes coincide, the prefix and suffix recovery sets remain disjoint.

For two requests arriving in the opposite order, a comparator first orders their prefixes, conditional swaps route the two suffixes to the prefix and suffix recovery branches, and a final swap restores the output order. This costs only $\text{poly}(n)$ gates.

The map $(f_0, \dots, f_{M-1}) \mapsto (g_0, \dots, g_M)$ is, up to an invertible change of information coordinates, the length- $(M+1)$ single-parity-check code. The two representations above are disjoint recovery sets for each information coordinate. To serve exponentially many requests, we use a constant-rate code with many geometric recovery sets.

5 The local-recovery gadget

We need a code with three features: it stores the 2^p prefix values using only $O(2^p)$ bits, it offers many recovery sets for every value, and the circuit can compute the addresses in those sets cheaply. The following product code supplies all three. [Proposition 5.2](#) is the interface used by the rest of the proof; the polynomial construction establishes that interface.

Fix an integer $\ell \geq 2$ and a field $\mathbb{F}_q$ of characteristic two, where $q = 2^b$. Set

$$s_0 = \left\lfloor \frac{q-1}{\ell} \right\rfloor$$

and fix a polynomial-basis representation of $\mathbb{F}_q$. If α_i denotes the field element whose canonical b -bit representation is the integer i , choose the efficiently indexed set

$$A = \{\alpha_0, \dots, \alpha_{s_0-1}\}.$$

Let $\mathcal{V}$ be the space of polynomials

$$P(X_1, \dots, X_\ell) \in \mathbb{F}_q[X_1, \dots, X_\ell]$$

having degree less than s_0 in every variable. Repeated univariate interpolation shows that evaluation on A^ℓ uniquely determines P , and

$$\deg P \leq \ell(s_0 - 1) \leq q - 2.$$

The evaluation map

$$\text{Enc} : \mathbb{F}_q^{A^\ell} \longrightarrow \mathbb{F}_q^{\mathbb{F}_q^\ell}, \quad (P(u))_{u \in A^\ell} \longmapsto (P(y))_{y \in \mathbb{F}_q^\ell},$$

is a systematic linear code: the original information symbols appear unchanged at the coordinates in A^ℓ . Its length and dimension are

$$N = q^\ell, \quad K = s_0^\ell = \Theta_\ell(q^\ell).$$

5.1 Packing Boolean restrictions

Split an n -bit input as

$$x = (a, z), \quad |a| = p, \quad |z| = d = n - p.$$

For each fixed suffix z , pack the 2^p bits

$$(f(a, z))_{a \in \{0,1\}^p}$$

into K field symbols, using the fixed $\mathbb{F}_2$ -basis of $\mathbb{F}_q$ and padding unused bit positions with zeros. More explicitly, interpret a prefix a as an integer $I(a) \in [2^p]_0$, and set

$$s(a) = \left\lfloor \frac{I(a)}{b} \right\rfloor, \quad j(a) = I(a) \bmod b.$$

Write $s(a)$ in base s_0 using exactly ℓ digits $i_1(a), \dots, i_\ell(a) \in [s_0]_0$, and define

$$u(a) = (\alpha_{i_1(a)}, \dots, \alpha_{i_\ell(a)}) \in A^\ell.$$

Thus bit $j(a)$ of the information symbol at $u(a)$ is $f(a, z)$. Fixed-divisor division, remainder, and base conversion have Boolean circuits of size $\text{poly}_\ell(p, \log q)$, so computing $(u(a), j(a))$ for all requests costs $\tilde{O}_\ell(t)$ gates.

Choose q , among powers of two, minimally so that

$$Kb \geq 2^p.$$

For all sufficiently large p (depending on ℓ), the corresponding code dimensions at field sizes q and $q/2$ are both $\Theta_\ell(q^\ell)$. Minimality then gives

$$q^\ell b = \Theta_\ell(2^p). \tag{1}$$

Indeed, the lower bound follows from $Kb \leq q^\ell b$. For the upper bound, the preceding candidate $q/2$ has bit width $b-1$, fails the packing inequality, and has dimension $\Omega_\ell(q^\ell)$. Hence $q^\ell(b-1) = O_\ell(2^p)$; since $b \leq 2(b-1)$ for $b \geq 2$, the upper bound in (1) follows. Equivalently,

$$\ell \log_2 q + \log_2 \log_2 q = p + O_\ell(1), \quad \log_2 q = \frac{p}{\ell} + O_\ell(\log p). \tag{2}$$

Let P_z be the polynomial whose information symbols are the packed bits. For each code coordinate $y \in \mathbb{F}_q^\ell$, define the field-valued resource

$$G_y(z) = P_z(y).$$

Each of the b coordinate bits of G_y is a Boolean function on d variables, determined by f , y , and the chosen field basis. For $r \in [b]_0$, write

$$H_{y,r}(z) = (G_y(z))_r.$$

These definitions view the same array in two ways:

Held fixed	What varies	What the values describe
Suffix z	Code coordinate y	One codeword P_z
Code coordinate y	Suffix z	One resource function G_y

The decoder combines coordinates of a codeword. The resource circuit evaluates one column of this array at the suffix routed to it. No encoding circuit or full array is evaluated on the live inputs: the array specifies which shorter functions are synthesized.

5.2 Affine-line recovery

Lemma 5.1 (Line identity). *Let $P : \mathbb{F}_q^\ell \rightarrow \mathbb{F}_q$ be represented by a polynomial of total degree at most $q - 2$. For every $u \in \mathbb{F}_q^\ell$ and every nonzero direction $v \in \mathbb{F}_q^\ell$,*

$$P(u) = - \sum_{\lambda \in \mathbb{F}_q^*} P(u + \lambda v).$$

In characteristic two, the minus sign may be omitted.

Proof. The restriction $Q(\lambda) = P(u + \lambda v)$ is a univariate polynomial of degree at most $q - 2$. The sum of the constant monomial over $\mathbb{F}_q$ is $q \cdot 1 = 0$ in characteristic two. For $1 \leq j \leq q - 2$, the sum of λ^j over $\mathbb{F}_q$ vanishes because j is not divisible by $q - 1$, and the zero term contributes nothing. Applying these identities monomial by monomial gives $\sum_{\lambda \in \mathbb{F}_q} Q(\lambda) = 0$. Separating the term $\lambda = 0$ proves the claim. $\square$

Thus every affine line through an information point u supplies a recovery set consisting of its $q - 1$ non-target points. We identify two nonzero direction vectors when one is a nonzero scalar multiple of the other and write $[v]$ for the resulting projective direction. The number of such directions through a point is

$$D_q = \frac{q^\ell - 1}{q - 1} = 1 + q + \dots + q^{\ell-1}, \quad q^{\ell-1} \leq D_q < 2q^{\ell-1}. \quad (3)$$

For fixed ℓ and $p \rightarrow \infty$, Equations (2) and (3) give the parameter dictionary

$$q^\ell = 2^{p+o(p)}, \quad q = 2^{p/\ell+o(p)}, \quad D_q = q^{\ell-1+o(1)} = 2^{((\ell-1)/\ell)p+o(p)}. \quad (4)$$

Two distinct lines through the same target intersect only at that target, which is absent from both recovery sets. Repeated requests for one information symbol therefore cause no additional difficulty. A later target may itself belong to the occupied set, but it blocks no direction because every new recovery set omits its own target.

The facts needed later can now be packaged without reference to the internal polynomial representation.

Proposition 5.2 (Local-recovery gadget). *Fix $\ell \geq 2$ and a split $n = p + d$, with p sufficiently large depending on ℓ , and choose $q = 2^b$ by the packing rule above. For every $f : \{0, 1\}^n \rightarrow \{0, 1\}$ there are $q^\ell b$ Boolean resource functions $H_{y,r} : \{0, 1\}^d \rightarrow \{0, 1\}$ with the following properties.*

- (i) *The resource count satisfies $q^\ell b = \Theta_\ell(2^p)$.*
- (ii) *A prefix a determines an information point $u(a) \in A^\ell$ and a bit position $j(a) \in [b]_0$ by a circuit of size $\text{poly}_\ell(p, \log q)$.*
- (iii) *For every projective direction $[v]$, the set*

$$R(u(a), [v]) = \{u(a) + \lambda v : \lambda \in \mathbb{F}_q^*\}$$

has $q - 1$ coordinates and recovers the requested bit by

$$f(a, z) = \bigoplus_{y \in R(u(a), [v])} H_{y, j(a)}(z).$$

There are $D_q = (q^\ell - 1)/(q - 1)$ such recovery sets.

(iv) If $U \subseteq \mathbb{F}_q^\ell$ is already occupied, at most $|U \setminus \{u(a)\}|$ directions have $R(u(a), [v]) \cap U \neq \emptyset$.

Proof. The packing construction gives (i) and (ii). The line identity, followed by selection of coordinate bit $j(a)$ in the fixed $\mathbb{F}_2$ basis, gives (iii). For (iv), every $y \in U \setminus \{u(a)\}$ lies on exactly one projective line through $u(a)$, while the omitted target $u(a)$ blocks none. $\square$

Thus one desired bit has become $q - 1$ requests to shorter resource functions, but those coordinates may be chosen from any of D_q recovery sets. The next section turns the blocking guarantee in (iv) into a deterministic circuit and then routes only the coordinates actually selected.

6 Deterministic scheduling and routing

The scheduler maps target coordinates to disjoint recovery sets. A greedy construction first shows how the blocking bound gives a circuit of size $\tilde{O}_\ell(g^2q)$ for g requests. Fixed menus reduce this cost to $\tilde{O}_\ell(gq)$. Routing then connects the selected addresses to the resource circuits. The main proof uses the menu scheduler; the greedy version supplies the explicit recursive alternative in [Appendix A](#).

6.1 Greedy line scheduling

The geometric count says that a suitable line exists, but a circuit must find one from the target points. We do this greedily. Previously occupied points rule out directions; projective ranking and sorting let an ordinary Boolean circuit choose the first direction that remains.

Consider a multiset of g requested information points $u_1, \dots, u_g \in A^\ell$. Process the requests in order. Suppose disjoint recovery sets have been selected for requests $1, \dots, j - 1$, and let U be their union. Then

$$|U| = (j - 1)(q - 1) < gq.$$

For the target u_j , every point $y \in U \setminus \{u_j\}$ forbids at most one projective direction, namely the direction of $y - u_j$. The point u_j itself forbids none because it is omitted from the new recovery set.

Lemma 6.1 (Projective indexing). *Projective directions in $\mathbb{F}_q^\ell$ have rank and unrank circuits of size $\text{poly}_\ell(b)$, with ranks in $[D_q]_0$.*

Implementation idea. The indexing normalizes the first nonzero coordinate to one and orders the normalized vectors by the position of that coordinate and then by their remaining base- q digits. The explicit circuits are given in [Appendix B.1](#).

Lemma 6.2 (Greedy line scheduler). *If*

$$gq < D_q, \tag{5}$$

then every multiset of g target points admits pairwise disjoint affine-line recovery sets. Given the targets, the recovery sets can be selected by a Boolean circuit of size

$$\tilde{O}_\ell(g^2q).$$

Proof idea. At stage j , the blocking property in [Proposition 5.2](#) leaves a valid direction because fewer than D_q directions are forbidden. The circuit ranks the forbidden directions, sorts those ranks, and selects the least missing one; stage j costs $\tilde{O}_\ell(jq)$. The treatment of duplicate targets, sentinel ranks, and the least-missing-rank circuit is given in [Appendix B.2](#). Summing over the stages gives the stated cost.

6.2 A nonuniform scheduler with linear dependence on the batch size

The quadratic scan can be replaced by phases that accept half the remaining requests. We prove that a small fixed menu of candidate directions works for every possible phase state, then give a circuit that tests the menu. The existence proof uses randomness only to choose constants of the circuit; its evaluation is deterministic on every input.

There are three steps. First, one random direction assignment succeeds with very high probability for any fixed state. Second, a union bound supplies a short menu that contains a successful assignment for every state. Third, a sorting circuit finds such an assignment without multiplying the occupied-set cost by the menu length. Here g is the original batch size, k is the number of pending requests, and U contains the points reserved earlier. A menu entry is a tuple of k directions, one for each pending request.

Lemma 6.3 (A partial schedule with exponentially small failure). *Put $N = q^\ell$. Suppose $512gq \leq D_q$, let $1 \leq k \leq g$, and let U be the union of at most g affine-line recovery sets. Fix any k targets, and choose one projective direction independently and uniformly for each target. Call a candidate recovery set clean if it avoids U and every other candidate recovery set. Then*

$$\Pr\{\text{fewer than } \lceil k/2 \rceil \text{ sets are clean}\} \leq 2^{-k}.$$

Proof. For any fixed set B , a random recovery set through any fixed target meets B with probability at most $|B|/D_q$: each point other than the target blocks exactly one direction. Consequently, the probability of meeting U is at most $a_0 = g(q-1)/D_q$, and that of meeting another fixed candidate recovery set is at most $a_1 = (q-1)/D_q$.

Pairwise collision events are dependent, so we cannot multiply these bounds directly. Instead, we extract a set of bad requests whose collision witnesses all lie outside that set, and condition on the outside directions.

Form a graph on $\{0, 1, \dots, k\}$. Join a request to vertex 0 when its recovery set meets U , and join two requests when their recovery sets meet. The nonclean requests are exactly the nonisolated vertices other than 0. Take a spanning forest of the nontrivial components and two-color it. Every nonclean request has a forest neighbor of the opposite color. If at least $\lceil k/2 \rceil$ requests are nonclean, one color contains at least $h = \lceil k/4 \rceil$ of them. Choose exactly h such requests as a set S , excluding vertex 0. Every request in S then has a collision witness outside S , either vertex 0 or an unselected request.

For each fixed set S of size h , condition on all directions outside S . Their recovery sets, together with U , form a fixed set of at most $(g+k)(q-1)$ points. Each selected request meets that set with probability at most $a_0 + ka_1$. These h tests are conditionally independent, since each uses only its own unexposed direction. Averaging over the complementary directions and taking a union bound

over S gives

$$\Pr\{\text{such a collision cut exists}\} \leq \binom{k}{h} (a_0 + ka_1)^h \leq 2^k (1/256)^{k/4} = 2^{-k}.$$

Here $a_0 + ka_1 \leq 2gq/D_q \leq 1/256$. Failure of the stated progress condition implies at least $\lceil k/2 \rceil$ nonclean requests, so this proves the claim. $\square$

Lemma 6.4 (Universal menus). *Under the same slack condition, for each $1 \leq k \leq g$ there is a fixed menu of*

$$r_k = \left\lceil \frac{g(1 + 3\log_2 N) + 1}{k} \right\rceil \quad (6)$$

direction k -tuples such that every set U described in Lemma 6.3 and every ordered list of k targets has a menu entry with at least $\lceil k/2 \rceil$ clean recovery sets.

Proof. A recovery set is specified by an anchor and a direction, giving at most ND_q descriptions. Allowing an unused marker in each of g slots, the number of occupied-set descriptions is at most $(1 + ND_q)^g$. The targets have at most N^k descriptions. Since $D_q \leq N$ and $k \leq g$, the total number of states is at most

$$(1 + ND_q)^g N^k \leq 2^{g(1+3\log_2 N)}.$$

Counting line descriptions is essential: treating U as an arbitrary list of $g(q-1)$ points would introduce an extra factor of q in this exponent. Choose r_k candidate tuples independently. For a fixed state, the probability that every tuple fails is at most 2^{-kr_k} by Lemma 6.3. A union bound over states is strictly below one by (6). Fix a menu working for all states as part of the circuit for parameters g, k, q, ℓ . $\square$

Theorem 6.5 (Nonuniform linear scheduling). *If $512gq \leq D_q$, a deterministic nonuniform Boolean circuit maps every ordered multiset of g targets in $\mathbb{F}_q^\ell$ to enumerated pairwise disjoint affine-line recovery sets using $\tilde{O}_\ell(gq)$ gates. More explicitly, its size is $O(gq\Lambda^5)$ for $\Lambda = \ell \log_2 q + \log_2(g+2)$. The circuit's fixed menus have total description length $O(g(\log N)^2 \log(g+2))$ bits.*

Proof. Maintain k active targets and the occupied set from the $g-k$ accepted requests. Initially $k = g$ and U is empty. Test the menu from Lemma 6.4, choose its first successful entry, and accept exactly the first $\lceil k/2 \rceil$ clean requests in that entry. These sets avoid U and each other. Compact the other $\lceil k/2 \rceil$ requests, retaining their original identifiers. This gives a fixed sequence of at most $1 + \lfloor \log_2 g \rfloor$ phase sizes, and never occupies more than g recovery sets. After the last phase, sort accepted records by original identifier to enumerate the answer sets in request order.

It remains to bound the cost of testing a menu. Generate records for all $r_k k(q-1)$ candidate points, labelled by candidate, request, and line position, and append the $(g-k)(q-1)$ occupied points once. Since $r_k k = O(g \log N)$, there are $O(gq \log N)$ records. Use these fixed passes:

- (i) Sort by point, with occupied records first at equal points. Along each equal-point run propagate an occupancy bit. In sorted position i , the update is $e_i = \text{occupied}_i \vee ((y_i = y_{i-1}) \wedge e_{i-1})$, with a dummy predecessor at the boundary. Thus every candidate point learns whether it lies in U .
- (ii) Compact candidate records and sort them by (candidate, y). Mark a record when its predecessor or successor has the same key. Points on one recovery line are distinct, so such a match is exactly an intersection between two requests in the same candidate.

- (iii) Sort by (candidate, request, line position) and OR the $q - 1$ bad-point flags for each request. Count the clean requests in each candidate and use priority selection to choose its first successful entry. Fixed multiplexers and compaction implement the acceptance step.

Every record has polynomial width in $\ell \log q$ and $\log(g + 2)$, and a constant number of sorting networks and linear scans suffices. Thus a phase has size $O(gq\Lambda^4)$: there are $O(gq\Lambda)$ records, with $O(\Lambda)$ bits per point record and $O(\Lambda^2)$ sorting-network comparators per record. Candidate directions and line scalars are hardwired, so generating a candidate point adds a constant field vector to the target using $O(\ell \log q)$ gates. Selecting and compacting whole line records costs at most $O(gq\Lambda^3)$ additional gates. There are $O(\Lambda)$ phases, proving the explicit bound. The menus contain $O(g \log N)$ directions per phase, each of $O(\log N)$ bits, giving the description bound. $\square$

6.3 Scatter and gather

The scheduler produces a sparse list of resource addresses, whereas the resource circuits occupy fixed positions in the circuit. *Scatter* sends each requested suffix to its resource position; *gather* returns the resulting value to the request that needs it. Sorting implements both maps with cost proportional, up to polynomial factors, to the lists being connected. With globally disjoint recovery sets, each resource position receives at most one suffix.

Every non-target point of a selected line produces an incidence record containing a request identifier, a line position, the point's code coordinate y , and the suffix to be evaluated. There are exactly $t(q - 1) < tq$ such records, with distinct code coordinates.

The routing consists of four fixed sorting-and-local-update passes. Here “key” is the resource coordinate y ; any unlisted tie is broken by the record's original position. The record-type order is pass-specific and is shown explicitly in the table.

Pass	Records	Sort order	Local result
Scatter match	incidence, slot	(key, type); incidence before slot	a matched slot takes its predecessor's suffix; otherwise it takes a dummy
Scatter order	incidence, filled slot	(type, key); slots first	slots are in canonical key order
Gather match	value, saved incidence	(key, type); value before incidence	a matched incidence takes its predecessor's value
Gather order	value, updated incidence	(type, request, line position); incidences first	decoder inputs are in canonical request order

Scatter records carry a d -bit suffix. After resource evaluation, value records carry the b -bit field value $G_y(z)$. Both payloads fit within the polynomial record-width bound.

Lemma 6.6 (Record routing). *Let a fixed table contain M distinct keys. Suppose every incidence key belongs to this table and occurs in at most one of the $t(q - 1)$ incidence records. With keys and payloads of width polynomial in n , $\ell \log q$, $\log(t + 2)$, and $\log(M + 2)$, suffixes can be scattered to their keyed slots and resource values gathered back to their request records using circuits of total size*

$$\tilde{O}_\ell(tq + M).$$

Implementation idea. The four passes in the table use a constant number of sorting networks on

$O(tq + M)$ records of polynomial width. The exact fixed-wire treatment of unmatched records and array boundaries is given in [Appendix B.3](#).

For the resource bank above, the table consists of the $M = q^\ell$ field points.

7 The composition bound

Disjoint recovery lets each resource circuit serve at most one suffix. There are $q^\ell b$ Boolean resources, each on d variables, so ordinary one-copy synthesis bounds their total size by $q^\ell bL(d)$. The remaining cost comes from selecting recovery sets, routing suffixes and values, and XOR decoding.

Proposition 7.1 (Scheduler-sensitive composition). *Fix $\ell \geq 2$ and the local-recovery gadget for a split $n = p + d$. Suppose a Boolean circuit of size at most $S(t, q)$ maps every ordered list of t information points to enumerated pairwise disjoint affine-line recovery sets. Then every $f : \{0, 1\}^n \rightarrow \{0, 1\}$ satisfies*

$$C(f^{\times t}) \leq \underbrace{q^\ell bL(d)}_{\text{resource circuits}} + \underbrace{S(t, q)}_{\text{schedule selection}} + \underbrace{\tilde{O}_\ell(tq + q^\ell)}_{\text{routing and decoding}}. \quad (7)$$

Proof. Map each prefix to its information point and bit position, and apply the scheduler. This costs $\tilde{O}_\ell(t) + S(t, q)$ gates. By [Lemma 6.6](#), scatter the assigned suffixes to the q^ℓ resource positions, filling unused positions with a fixed dummy suffix. Disjointness ensures that each position receives at most one requested suffix.

Evaluate the b Boolean functions $H_{y,r}$ at each position y , for a total of $q^\ell bL(d)$ gates, and gather the resulting field values. Routing costs $\tilde{O}_\ell(tq + q^\ell)$. For each request, XOR its $q - 1$ returned values and select the requested bit. This is correct by [Lemma 5.1](#) and costs $O(tqb)$, which the displayed overhead absorbs. $\square$

8 Direct proof of exponential-range mass production

The parameter choice balances storage against recovery capacity. Keeping $d \approx \delta n$ suffix bits leaves about $2^{(1-\delta)n}$ encoded prefix bits. Since a recovery set occupies about q points while a target has about $q^{\ell-1}$ directions, the scheduler needs t to be smaller than a constant multiple of $q^{\ell-2}$. The inequality below expresses exactly this requirement at the level of exponential rates.

Direct proof of [Theorem 1.1](#). Fix $\gamma < 1$, choose any fixed $0 < \delta < 1 - \gamma$, and put $d = \lceil \delta n \rceil$ and $p = n - d$. Choose a fixed $\ell \geq 2$ large enough that

$$\gamma < (1 - 2/\ell)(1 - \delta). \quad (8)$$

Use the product-code gadget with its minimal packing field. Its parameters satisfy

$$\log_2 q = (1 - \delta)n/\ell + o(n), \quad \log_2 D_q = (1 - 1/\ell)(1 - \delta)n + o(n).$$

Thus (8) implies $512tq \leq D_q$ for all sufficiently large n and every $t \leq 2^{\gamma n}$. Apply [Theorem 6.5](#) and [Proposition 7.1](#). This gives

$$C(f^{\times t}) \leq q^\ell bL(d) + \tilde{O}_\ell(tq + q^\ell).$$

The first term is $O_\ell(2^p 2^d/d) = O_\gamma(2^n/n)$. The two overhead exponents are at most $\gamma + (1 - \delta)/\ell < 1$ and $1 - \delta < 1$, respectively, with fixed positive margins. All suppressed polynomial factors are therefore negligible compared with $2^n/n$. Enlarge the constant to cover the finitely many initial lengths by ordinary circuit replication. $\square$

9 High-rate codes and the leading coefficient

The preceding proof has two constant losses. First, the product code's rate approaches $\ell^{-\ell}$: it stores substantially more symbols than the information it carries. Second, rounding up to the next allowed field size can leave a constant fraction of the information positions unused. We remove the first loss with a code of rate tending to one and the second with several smaller codewords, so that only the last one needs padding. The scheduler and decoder continue to use the same line-recovery interface.

Lifted Reed–Solomon codes already provide rate tending to one with line-parity recovery [GKS13, HPPV20]. We give an elementary sufficient monomial construction below, followed by the packing and circuit-cost accounting.

Lemma 9.1 (High-rate line-parity code). *Fix integers $\ell \geq 2$ and $h \geq \lceil \log_2 \ell \rceil$. Put $q = 2^b$ and $N = q^\ell$. There is a linear code on $\mathbb{F}_q^\ell$ of dimension K such that*

$$\frac{K}{N} \geq 1 - (1 - 2^{-\ell h})^{\lfloor b/h \rfloor} = 1 - o_{\ell,h}(1) \quad (b \rightarrow \infty), \quad (9)$$

and each word P satisfies

$$P(u) = \sum_{\lambda \in \mathbb{F}_q^*} P(u + \lambda v)$$

for every target u and every nonzero direction v . The code can be made systematic on a fixed set of K evaluation points.

Proof. The low-degree condition in Lemma 5.1 was sufficient for line parity, but stronger than necessary. For the sum over $\mathbb{F}_q$ to vanish, it is enough that every positive degree appearing on a line avoid multiples of $q - 1$. We enforce this weaker condition using a common block of zero binary digits.

Consider reduced monomials $X_1^{e_1} \cdots X_\ell^{e_\ell}$ with $0 \leq e_i < 2^b$. Partition the first $h \lfloor b/h \rfloor$ binary digit positions into disjoint blocks of length h . Retain a monomial if some block is zero in *all* of its exponents. The fraction retained is exactly $1 - (1 - 2^{-\ell h})^{\lfloor b/h \rfloor}$. These monomials are linearly independent as evaluation functions, by repeated univariate interpolation.

To check line parity, expand a retained monomial at $u + \lambda v$. In characteristic two, every possibly nonzero term in this expansion has λ -degree $J = \sum_i j_i$, where the one-bits of j_i are a subset of those of e_i . This follows directly by writing $(u_i + \lambda v_i)^{e_i}$ as a product over the one-bits of e_i and using $(a + c)^{2^r} = a^{2^r} + c^{2^r}$.

Suppose the common zero block starts at position s . Every j_i has the same zero block, so its residue modulo 2^{s+h} is at most $2^s - 1$. Consequently,

$$J \bmod 2^{s+h} = \sum_i (j_i \bmod 2^{s+h}) \leq \ell(2^s - 1) \leq 2^{s+h} - \ell.$$

There is no wraparound in this equality. If J were a positive multiple $a(2^b - 1)$, we would have $1 \leq a \leq \ell - 1$: the zero block makes every $j_i < 2^b - 1$. Since $s + h \leq b$ and $a < 2^{s+h}$, its residue would instead be $2^{s+h} - a \geq 2^{s+h} - \ell + 1$, a contradiction. Thus every positive λ -degree is not divisible by $q - 1$. The monomial-sum calculation in [Lemma 5.1](#) gives zero for the sum over $\mathbb{F}_q$, including the constant term. Linearity proves line parity for the span of the retained monomials.

Finally, choose K independent coordinate columns of a generator matrix. Evaluation on those coordinates is an isomorphism to $\mathbb{F}_q^K$, giving a systematic information set. The choice and any change of basis are offline parts of the nonuniform code specification. $\square$

To index the information set, store its ordered list of K points as a hardwired table, and sort its records together with all t live symbol indices, placing a table record before its queries at an equal index. Propagate the table value along each equal-index run, then sort queries back to request order. This costs $\tilde{O}_\ell(K + t)$ gates and handles repeated indices. It is the same batched lookup primitive used for occupancy broadcast, with a point as payload.

Proof of Theorem 1.2. Fix $0 < \delta < 1 - \gamma$, put $d = \lceil \delta n \rceil$, $p = n - d$, and choose a fixed ℓ satisfying (8). Fix $h = \lceil \log_2 \ell \rceil$ in [Lemma 9.1](#). Instead of the minimal packing field, take

$$b = \left\lfloor \frac{p - 3 \log_2(n + 2)}{\ell} \right\rfloor, \quad q = 2^b, \quad B = \left\lceil \frac{2^p}{Kb} \right\rceil,$$

for sufficiently large n , where K is the dimension in [Lemma 9.1](#). Use B separate copies of this code. Each has Kb information bits, so they hold the 2^p prefix values with only the last code's information positions padded. Because $K/N \rightarrow 1$ and $Nb = o(2^p)$, their total number of Boolean resources is

$$2^p \leq BNb \leq \frac{N}{K} 2^p + Nb = (1 + o(1))2^p. \quad (10)$$

Here $B = \text{poly}_\ell(n)$ and $\log_2 q = (1 - \delta)n/\ell + O_\ell(\log n)$.

Fixed-divisor arithmetic maps each prefix to its code number, information index, and bit position. The batched table lookup above supplies all target points. Schedule these t targets together, even when they refer to different code copies. By [Theorem 6.5](#), the resulting recovery sets are globally disjoint. Attaching the requested code number to each incidence therefore gives unique keys (code, y). Route the $t(q - 1)$ incidences against BN slots by the four-pass construction of [Lemma 6.6](#).

For each of the BNb Boolean resource functions on d variables, use the sharp one-copy synthesis bound $L(d) \leq (1 + o(1))2^d/d$ in the stated basis [\[FM05\]](#). The complete bound is

$$C(f^{\times t}) \leq BNbL(d) + \tilde{O}_\ell(tq + BN + K) \leq \left(\frac{1}{\delta} + o(1) \right) \frac{2^n}{n}.$$

Indeed, (10) controls the resource term; the other terms have exponents at most $\gamma + (1 - \delta)/\ell < 1$ and $1 - \delta < 1$. All estimates are uniform over f and the allowed t .

For every fixed $\delta < 1 - \gamma$ we have proved a limiting upper coefficient $1/\delta$. Taking the infimum of these bounds as δ increases to $1 - \gamma$ proves the theorem. This last step takes a limit of fixed-parameter bounds; it makes no assertion uniform in growing ℓ or in $\gamma \rightarrow 1$. $\square$

10 Linear depth at the same size order

To prove [Theorem 1.3](#), we return to the constant-rate product code of [Section 5](#) and adapt its packing to make the prefix map a wiring operation. The scheduler also needs a depth refinement: halving uses $O(\log t)$ dependent phases. Testing larger menus lets each phase leave only an exponentially small fraction of requests unfinished. A constant number of phases then suffices, with total gate count still negligible beside $2^n/n$.

The collision estimate below refines the same forest-conditioning argument used in [Lemma 6.3](#). To implement a phase at low depth we use Holmgren and Rothblum's sorting theorem [[HR24](#), Lemma 6]: R records of w bits, with κ -bit keys, can be sorted by a bounded-fan-in Boolean circuit of size $O(Rw \log(R+2))$ and depth $O(\kappa + \log(R+2))$. Their proof uses the sorting networks of Ajtai, Komlós, and Szemerédi, with a bit-position argument that bounds the depth of the complete Boolean network [[HR24](#), Section 4].

10.1 Menus that leave few requests unfinished

Lemma 10.1 (Collision tail). *Let $1 \leq k \leq g$, let U be a union of at most g recovery sets, and fix any ordered list of k target points. Choose their directions independently and uniformly. A request is bad if its candidate recovery set meets U or another candidate recovery set. For every integer $0 \leq v < k$, put $h = \lceil (v+1)/2 \rceil$. Then*

$$\Pr\{\text{more than } v \text{ requests are bad}\} \leq \binom{k}{h} \left(\frac{2gq}{D_q} \right)^h.$$

Proof. Form the collision graph on the requests and a vertex 0 representing U , as in [Lemma 6.3](#). If more than v requests are bad, two-color a spanning forest of its nontrivial components. One color contains at least h bad requests, not counting vertex 0. Choose exactly h of them as S . Every member of S has a forest neighbor outside S , which may be vertex 0.

For a fixed S , condition on all directions outside S . Their recovery sets together with U contain at most $2g(q-1)$ points. Each target in S independently chooses a direction whose recovery set meets this fixed union with probability at most $2gq/D_q$. A union bound over the $\binom{k}{h}$ possible choices of S gives the estimate. $\square$

Lemma 10.2 (Menus with a prescribed remainder). *Put $N = q^\ell$ and $E_g = g(1 + 3 \log_2 N)$. Suppose $0 < \alpha \leq 1/2$ and*

$$\frac{4e g q}{D_q} \leq \alpha^2,$$

where e is the base of the natural logarithm. For each $1 \leq k \leq g$, define

$$v_k = \lfloor \alpha k \rfloor, \quad h_k = \left\lceil \frac{v_k + 1}{2} \right\rceil, \quad r_k = \left\lceil \frac{E_g + 1}{h_k \log_2(1/\alpha)} \right\rceil.$$

There is a fixed menu of r_k direction k -tuples such that, for every occupied set U that is a union of at most g recovery sets and every ordered list of k targets, a menu entry has at least $k - v_k$ clean recovery sets. Moreover,

$$r_k k \leq g + \frac{2(E_g + 1)}{\alpha \log_2(1/\alpha)}. \quad (11)$$

Proof. We have $h_k > \alpha k/2$. By [Lemma 10.1](#) and the inequality $\binom{k}{h} \leq (ek/h)^h$, one random tuple fails with probability at most

$$\left(\frac{ek}{h_k} \frac{2gq}{D_q}\right)^{h_k} \leq \left(\frac{4e}{\alpha} \frac{gq}{D_q}\right)^{h_k} \leq \alpha^{h_k}.$$

As in [Lemma 6.4](#), there are at most $(1 + ND_q)^g N^k \leq 2^{E_g}$ states to serve. A menu of r_k independent tuples fails on any particular state with probability at most $2^{-r_k h_k \log_2(1/\alpha)} \leq 2^{-E_g - 1}$. The union bound is less than one, so a menu works for all states. Finally, the ceiling in r_k and $k/h_k < 2/\alpha$ give [\(11\)](#). $\square$

In a successful phase, accept exactly the first $k - v_k$ clean requests, reserve their recovery sets, and compact the remaining v_k requests. The successive values of k are fixed in advance, since each is replaced by $\lfloor \alpha k \rfloor$. The number of phases is at most

$$1 + \left\lfloor \frac{\log_2 g}{\log_2(1/\alpha)} \right\rfloor.$$

This includes the final phase, when $\lfloor \alpha k \rfloor = 0$.

10.2 Depth of one phase and of the resources

For a phase with k pending requests, generate the $r_k k(q - 1)$ candidate-point records and append the occupied points once. Thus the record count is

$$R = O(q(g + r_k k)).$$

The three verification passes from [Theorem 6.5](#) still apply. Candidate directions and line scalars are hardwired, so each candidate point is its target plus a fixed field vector.

The occupancy propagation in the first pass need not be a serial scan. Writing its update as $e_i = a_i \vee (b_i \wedge e_{i-1})$, encode it by the pair (a_i, b_i) . Composition is the associative operation

$$(a, b) \circ (c, d) = (a \vee (b \wedge c), b \wedge d).$$

A doubling prefix circuit therefore computes every occupancy bit in $O(\log(R + 2))$ depth and $O(R \log(R + 2))$ gates. This is the usual parallel-prefix method; a related propagation primitive is given in [\[HR24, Lemma 26\]](#), with credit there to Ladner and Fischer.

OR the bad-point flags along each request's line with a balanced tree. To test whether a candidate has at least $k - v_k$ clean requests, sort its k clean bits and inspect the appropriate fixed position. Prefix OR selects the first successful candidate. One-hot multiplexers select its records, and sorting by the clean flag and request position accepts exactly the required number of clean requests. All these operations have depth $O(w + \log(R + 2))$ and size $R \text{ poly}(w + \log(R + 2))$, when all keys and records have width $O(w)$. The record sorting steps use [\[HR24, Lemma 6\]](#). In the parameter choice below, $w = O_\gamma(n)$ and $\log(R + 2) = O_\gamma(n)$, so a phase has depth $O_\gamma(n)$.

The resource circuits run in parallel, so their contribution to depth is the maximum depth of any one resource. The following synthesis bound keeps this part linear as well.

Lemma 10.3 (One-copy synthesis with a depth bound). *Every Boolean function on $d \geq 1$ bits has a circuit of size $O(2^d/d)$ and depth $d + O(\log(d + 2))$.*

Proof. For $d \geq 4$, let $k = \lfloor \log_2 d \rfloor - 1$ and split the input into $d - k$ prefix bits and k suffix bits. Compute all minterms of the prefix with $O(2^{d-k})$ gates and depth $O(\log(d+2))$: recursively form the minterms on the two halves and AND every pair of half-minterms.

On the suffix, compute all 2^{2^k} Boolean functions. Computing all suffix minterms once and then the OR for each truth table uses $O(2^k 2^{2^k})$ gates and depth $k + O(\log(k+2))$. For each prefix assignment, select by fixed wiring the suffix function specified by the desired restriction. AND it with that prefix's minterm, and OR all 2^{d-k} terms using a balanced tree. The size is

$$O(2^{d-k} + 2^k 2^{2^k}) = O(2^d/d),$$

because $d/4 < 2^k \leq d/2$. The final OR contributes $d - k$ layers, giving the stated depth. Supplying negated input bits adds at most one layer. The finitely many smaller values of d are absorbed in the constants. $\square$

10.3 The simultaneous bound

Proof of Theorem 1.3. Fix $0 \leq \gamma < 1$, choose $0 < \delta < 1 - \gamma$, and take a fixed ℓ sufficiently large that

$$\gamma < (1 - 2/\ell)(1 - \delta).$$

Put $d = \lceil \delta n \rceil$, $p = n - d$, and $\lambda = (1 - \delta)/\ell$. We use the product-code construction, with a minor packing choice that makes the prefix map a wiring operation.

Let $c = \lceil \log_2 \ell \rceil + 1$. For a sufficiently large field $q = 2^b$, take A to be the first 2^{b-c} elements in the fixed binary representation of $\mathbb{F}_q$. Use the product grid A^ℓ , of size $K = 2^{\ell(b-c)}$. The interpolating polynomial has degree at most $\ell(2^{b-c} - 1) \leq q - 2$, so the same line-parity recovery identity applies. Pack only $b' = 2^{\lfloor \log_2 b \rfloor}$ information bits into each field symbol, padding its remaining $b - b'$ bits with zeros. Choose the least sufficiently large b such that $Kb' \geq 2^p$.

Since Kb' is a power of two, pad a prefix with leading zero bits and split it into ℓ coordinate indices of $b - c$ bits and one bit-position index of $\log_2 b'$ bits. Prepending c zeros to each coordinate index gives its field representation. Thus no division or table lookup is needed for this prefix map. Increasing b by one multiplies Kb' by at most $2^{\ell+1}$, while $b < 2b'$. Minimality therefore gives

$$q^\ell b = O_\ell(2^p), \quad \log_2 q = \lambda n + O_\ell(\log(n+2)).$$

The constant loss in this packing is harmless for the order-of-growth bound.

Set

$$\beta = (\ell - 2)\lambda - \gamma > 0, \quad \rho = \beta/4, \quad \alpha = 2^{-\lfloor \rho n \rfloor}.$$

For all sufficiently large n and uniformly for $g = t \leq 2^{\gamma n}$,

$$\frac{gq}{D_q} \leq 2^{-\beta n + O_\ell(\log(n+2))},$$

so $4e gq/D_q \leq \alpha^2$ and $0 < \alpha \leq 1/2$. Apply Lemma 10.2. Since $E_g = O_\gamma(gn)$ and $\log_2(1/\alpha) = \Theta_\gamma(n)$, (11) gives $r_k k = O_\gamma(g/\alpha)$ uniformly over every phase. The number of phases is $O_\gamma(1)$, and each phase uses

$$R \leq C_\gamma gq/\alpha \leq 2^{(\gamma+\lambda+\rho)n + O_\gamma(\log(n+2))}$$

records per phase, for a suitable constant C_γ . Moreover,

$$\gamma + \lambda + \rho = \frac{3}{4}\gamma + \frac{\ell+2}{4}\lambda < (\ell-1)\lambda < 1.$$

The scheduler consequently has depth $O_\gamma(n)$ and size $o_\gamma(2^n/n)$, even with the polynomial factors from its Boolean implementation. Its menus remain fixed nonuniform data.

Scatter and gather use the four passes of [Lemma 6.6](#), now implemented with Holmgren–Rothblum sorting. There are $O(gq + q^\ell)$ records of $O_\gamma(n)$ bits. Each pass has depth $O_\gamma(n)$, and their total size is $o_\gamma(2^n/n)$ because $\gamma + \lambda < 1$ and $\ell\lambda = 1 - \delta < 1$. Balanced XOR trees recover each field symbol in $O(\log q)$ depth; selecting its requested bit adds $O(\log(b+2))$ depth.

Finally, apply [Lemma 10.3](#) independently to the $q^\ell b$ Boolean resource functions. Their total size is

$$O\left(q^\ell b \frac{2^d}{d}\right) = O_\gamma(2^n/n),$$

and they run in parallel at depth $O_\gamma(n)$. The prefix map, the constant number of scheduling phases, routing, resource evaluation, and decoding therefore have total depth $O_\gamma(n)$ and total size $O_\gamma(2^n/n)$. Enlarge both constants to handle the finitely many initial input lengths. $\square$

The power-of-two packing and [Lemma 10.3](#) allow constant losses, so this proof controls the size order without retaining the leading coefficient of [Theorem 1.2](#).

10.4 Randomized construction from the truth table

Amplifying the menu union bound turns the preceding existence proof into a randomized construction.

Corollary 10.4 (Randomized construction). *For every fixed $0 \leq \gamma < 1$, there is a randomized algorithm that, given the truth table of $f : \{0, 1\}^n \rightarrow \{0, 1\}$ and an integer $1 \leq t \leq 2^{\gamma n}$, runs in time polynomial in 2^n and outputs a circuit of size $O_\gamma(2^n/n)$ and depth $O_\gamma(n)$. With probability at least $1 - 2^{-n}$ over the construction, that circuit computes $f^{\times t}$ correctly on every input batch.*

Proof. Choose a rational rate $\gamma < \hat{\gamma} < 1$ and use the proof of [Theorem 1.3](#) at rate $\hat{\gamma}$, also choosing rational δ . Its fixed numerical parameters are then rational, and its bounds remain $O_\gamma(2^n/n)$ and $O_\gamma(n)$. Let $P = O_\gamma(1)$ be its number of phases. The successive pending request counts k are fixed before any input is supplied. In each phase, sample a menu of independent uniform direction tuples, replacing its length by

$$r'_k = \left\lceil \frac{E_g + n + \lceil \log_2 P \rceil + 2}{h_k \log_2(1/\alpha)} \right\rceil.$$

The proof of [Lemma 10.2](#) bounds the probability that this menu fails on any state by

$$2^{E_g - r'_k h_k \log_2(1/\alpha)} \leq 2^{-n - \lceil \log_2 P \rceil - 2}.$$

A union bound over the phases is at most 2^{-n-2} . On the complementary event all menus work for every state, hence for every input batch and every choice of the resource truth tables. Since $E_g = \Theta_\gamma(gn)$, the enlarged menus still satisfy $r'_k k = O_\gamma(g/\alpha)$, preserving both circuit bounds.

All remaining construction steps take polynomial time in 2^n . The field has $b = O_\gamma(n)$ bits per element; exhaustive search and trial division find an irreducible polynomial of degree b in $2^{O_\gamma(n)}$

time. Product-grid interpolation constructs the resource truth tables in polynomially many field operations in their total table length, which is $2^{O_\gamma(n)}$. The synthesis in [Lemma 10.3](#) explicitly lists the required gates. The sorting circuits are log-space uniform [[HR24](#), Lemmas 8–9], and the other scheduling and routing gates are explicitly specified above.

For a worst-case polynomial running time, implement each uniform direction choice by rejection sampling an index in $[D_q]_0$. If L directions are needed in total, cap each choice at $n + \lceil \log_2(L+1) \rceil + 2$ trials. Each trial succeeds with probability at least $1/2$, so the probability of any exhausted cap is at most 2^{-n-2} . On that event output the all-zero circuit. Otherwise the chosen directions remain independent and uniform. Combining the two failure bounds proves the stated probability. The finitely many initial input lengths use elementary truth-table synthesis, with constants enlarged. $\square$

The output circuit uses no runtime randomness. The algorithm does not certify that its sampled menus work, and its time bound is polynomial in the full truth table, which may be exponentially larger than a succinct description of f .

11 Optimal approximate quantum state mass production

The upper bound in [Theorem 1.4](#) has three steps: evaluate Boolean functions coherently, use their outputs to select rotation angles, and assemble the target state from conditional rotations. The lower bound compares a packing of tensor-power states with the number of finite-gate circuits. Throughout, we charge every elementary gate and measure error on the entire batch.

11.1 Model and coherent Boolean evaluation

Fix the finite gate set $\mathcal{G} = \{H, T, T^\dagger, \text{CNOT}\}$, with arbitrary connectivity. Here H is the Hadamard gate, $T = \text{diag}(1, e^{i\pi/4})$, and CNOT is the controlled-NOT gate. Every gate has cost one; quantum depth counts layers of gates acting on pairwise disjoint qubits. Circuits have no measurements or postselection.

For state preparation, all qubits start in zero, the nt output qubits are fixed, and every scratch qubit returns exactly to zero. The number of scratch qubits is unrestricted, and the circuit may depend on the target state. The coherent subroutines below instead act on arbitrary input states, with the same requirement on scratch qubits.

For normalized pure states, write

$$d(\psi, \phi) = \sqrt{1 - |\langle \psi | \phi \rangle|^2}$$

for their trace distance; this identifies states differing only by a global phase. Let $Q_\varepsilon(\psi, t)$ be the minimum gate count whose output is within trace distance ε of $|\psi\rangle^{\otimes t}$, and set

$$M(n, t) = \max_{\psi \in \mathbb{C}^{2^n}, \|\psi\|=1} Q_{1/10}(\psi, t).$$

The construction below bounds these integer-valued minima uniformly, so the maximum is well-defined. The state is specified when the circuit is designed; no copy of it is supplied as an input.

We first implement Boolean mass production reversibly. The simulation also gives a depth bound for the resulting oracle.

Lemma 11.1 (Coherent Boolean mass production). *If a Boolean circuit of size s and depth D computes $f^{\times t}$, there is a quantum circuit of size $O(s + t + 1)$, depth $O((D + 1) \log(s + t + 2))$, and $O(s + t + 1)$ scratch qubits implementing*

$$|x_1, \dots, x_t\rangle |z_1, \dots, z_t\rangle |0^a\rangle \longmapsto |x_1, \dots, x_t\rangle |z_1 \oplus f(x_1), \dots, z_t \oplus f(x_t)\rangle |0^a\rangle.$$

In particular, for fixed $0 \leq \eta < 1$, $k \geq 1$, and $t \leq 2^{\eta k}$, this map has size $O_\eta(2^k/k + t)$ for every k -bit Boolean function f .

Proof. Delete gates outside the output cones. Assign a separate control qubit to each use of a Boolean wire, including its uses as a designated output. A wire with r uses supplies these controls by a balanced CNOT tree, using $O(r)$ gates and scratch qubits at depth $O(\log(r + 1))$. The total number of uses is at most $2s + t$, so this costs $O(s + t + 1)$ gates and scratch qubits in all.

Compute the Boolean gates in depth order, each into a fresh zero qubit. NOT, AND, and OR each require a constant number of NOT, CNOT, and Toffoli gates; repeated sources and constants are handled by their one-input or constant simplifications. These gates have exact constant-size implementations over $\mathcal{G}$. The separate control qubits make the simulations within a Boolean layer disjoint. After each layer, prepare its output fan-out trees in parallel. The initial input fan-out and the D gate layers therefore have total depth $O((D + 1) \log(s + t + 2))$.

Copy the designated output controls to the distinct z qubits in parallel, then reverse all gate computations and fan-out trees. Prepared constant sources and all scratch qubits are restored.

The displayed equality holds on every computational basis state, hence by linearity on every superposition and on inputs entangled with other registers. Apply [Theorem 1.1](#) for the final assertion. $\square$

Corollary 11.2 (Coherent evaluation at quadratic depth). *For fixed $0 \leq \gamma < 1$, every n -bit Boolean function and every $1 \leq t \leq 2^{\gamma n}$ admit the exact coherent evaluation map in [Lemma 11.1](#) with $O_\gamma(2^n/n)$ elementary gates, $O_\gamma(n^2)$ depth, and $O_\gamma(2^n/n)$ scratch qubits, all restored to zero.*

Proof. Apply [Lemma 11.1](#) to [Theorem 1.3](#), using $t = O_\gamma(2^n/n)$ and $\log(s + t + 2) = O_\gamma(n)$. $\square$

For state preparation we use the gate bound in [Lemma 11.1](#). The remaining rotation and synthesis steps below are analyzed for size only.

11.2 Conditional rotations with clean scratch space

For $a \in \{y, z\}$, let $R_a(\theta) = \exp(-i\theta\sigma_a/2)$, where σ_a is the corresponding Pauli matrix. A *multiplexed rotation* on k address qubits and one target qubit is

$$U_\theta = \sum_{x \in \{0,1\}^k} |x\rangle\langle x| \otimes R_a(\theta(x)).$$

Thus an address selects a rotation angle while remaining unchanged. We use $\|\cdot\|$ for the operator norm on operators.

Lemma 11.3 (Batched conditional rotations). *Fix $0 \leq \eta < 1$. Suppose $k \geq 1$, $t \leq 2^{\eta k}$, and $0 < \delta \leq 1/4$. Put $b = \lceil \log_2(8\pi/\delta) \rceil$. For every multiplexed rotation U_θ , a circuit over $\mathcal{G}$ with clean scratch space approximates $U_\theta^{\otimes t}$ to operator-norm error at most $t\delta$, using*

$$O_\eta \left(b \frac{2^k}{k} + t \text{ poly}(b) \right)$$

gates. The polynomial is fixed, independent of k, t, θ . The circuit preserves every address qubit exactly.

Proof. Represent each angle modulo 4π , and round it down to $4\pi \sum_{j=1}^b 2^{-j} f_j(x)$, where each f_j is a Boolean function of the address. Since $\|R_a(u) - R_a(v)\| \leq |u - v|/2$, this changes each selected rotation by at most $2\pi 2^{-b} \leq \delta/4$.

For each j , use [Lemma 11.1](#) to compute $f_j(x_1), \dots, f_j(x_t)$ into zero qubits. On block i , rotate the target by $4\pi 2^{-j}$ controlled by the computed bit, and then uncompute the bits. This charges $O_\eta(b 2^k/k + bt)$ gates for Boolean evaluation and its reversal. The address registers and all computed control bits are unchanged by the controlled rotations, so reversing the lookup erases its scratch space exactly.

It remains to synthesize a controlled rotation in the finite gate set without losing this property. Write X for the NOT matrix. Given a one-qubit word A approximating $R_a(\theta/2)$, use the two-CNOT circuit with block form

$$|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes AXA^\dagger X.$$

The identity $XR_a(u)X = R_a(-u)$ shows that the ideal choice of A implements the desired controlled rotation. For every implemented word A , the control-zero block is exactly I and the control is preserved exactly. A global phase in A cancels against $A^\dagger$. The Solovay–Kitaev theorem [\[DN06\]](#) supplies A of length polynomial in $\log(b/\delta)$ so that this controlled gate has error at most $\delta/(4b)$.

After lookup and uncomputation the resulting operation on the data is the tensor product of the corresponding approximate one-block multiplexors: each computed bit depends only on its own block’s address. The error in one block is at most $\delta/4 + b\delta/(4b) \leq \delta$. The tensor-product telescoping inequality gives error at most $t\delta$. There are bt synthesized controlled gates. Since $\log(b/\delta) = O(b)$, their total cost is $t \text{ poly}(b)$, as claimed. $\square$

Lemma 11.4 (State preparation by conditional rotations). *Up to global phase, every n -qubit pure state can be prepared using two multiplexed rotations, one about each of the y and z axes, at each address length $k = 0, \dots, n-1$.*

Proof. This is the standard recursive state-synthesis decomposition [\[SBM06, Kre23\]](#). To see it directly, write the target as

$$|\psi\rangle = \sum_{x \in \{0,1\}^{n-1}} |x\rangle (a_x|0\rangle + b_x|1\rangle).$$

For each x , choose angles θ_x, ϕ_x and a complex scalar c_x such that

$$a_x|0\rangle + b_x|1\rangle = c_x R_z(\phi_x) R_y(\theta_x) |0\rangle, \quad |c_x|^2 = |a_x|^2 + |b_x|^2.$$

Two rotations parametrize any normalized one-qubit state up to phase; when both amplitudes vanish take $c_x = 0$ and arbitrary angles. The state $\sum_x c_x |x\rangle$ is normalized. Prepare it recursively, adjoin a zero target, and apply the two rotations controlled by x . The induction ends with a scalar of modulus one, which contributes only an irrelevant global phase. $\square$

11.3 Upper bound and precision allocation

To control the error of the whole batch, we give each copy an $O(1/t)$ error budget. We allocate most of that budget to the largest address tables, which then need only $O(\log(t+1))$ angle bits. Smaller tables use finer precision at a geometrically smaller synthesis cost. Related allocations appear in [GKW26, Appendix B].

Proposition 11.5 (Quantum upper bound). *For every fixed $0 \leq \gamma < 1$, every n -qubit pure state ψ , and every $1 \leq t \leq 2^{\gamma n}$,*

$$Q_{1/10}(\psi, t) = O_\gamma \left(\frac{2^n}{n} \log_2(t+1) \right).$$

Proof. Fix constants $\gamma < \alpha < 1$ and $\gamma/\alpha < \eta < 1$, and put $m = \lceil \alpha n \rceil$ and $L = 1 + \log_2 t$. For sufficiently large n we have $1 \leq m < n$. Use Lemma 11.4 on each of the t blocks. At address length k , approximate each of the two rotations with the one-block error allowance

$$\delta_k = \frac{1}{40t} 2^{k-n}.$$

The number of angle bits in Lemma 11.3 is then

$$b_k = O(n - k + L) = O(n).$$

Since $t\delta_k$ bounds the error of each batched rotation, telescoping over the stages gives total operator-norm error at most

$$2t \sum_{k=0}^{n-1} \delta_k = \frac{1}{20} \sum_{k=0}^{n-1} 2^{k-n} < \frac{1}{10}.$$

Acting on the all-zero state gives the same bound on vector distance after aligning global phases, and hence on trace distance.

To apply the Boolean theorem at short address lengths, replace k by $k' = \max\{k, m\}$. When $k < m$, add zero address qubits and extend each angle function to ignore them. They are preserved and can be reused. The fixed rate η works at every stage, since

$$\log_2 t \leq \gamma n < \eta \alpha n \leq \eta k'.$$

Thus no stage invokes a mass-production bound outside its allowed range.

For $k \geq m$, Lemma 11.3 bounds the Boolean lookup cost, summed over the two rotations per stage, by

$$\begin{aligned} O_\gamma \left(\sum_{k=m}^{n-1} (n - k + L) \frac{2^k}{k} \right) &\leq O_\gamma \left(\frac{2^n}{n} \sum_{j \geq 1} (j + L) 2^{-j} \right) \\ &= O_\gamma \left(\frac{2^n}{n} L \right). \end{aligned}$$

Here $k \geq \alpha n$ bounds the denominators, and the two geometric sums are finite. The $O(n)$ short stages each have $b_k = O(n)$ and address length m , for a total lookup cost

$$O_\gamma(n^2 2^m / m) = O_\gamma(n 2^{\alpha n}) = o(2^n / n).$$

Finally, all rotation-synthesis and output-copying costs sum to $t \text{poly}(n) = o(2^n / n)$ uniformly for $t \leq 2^{\gamma n}$. The strict fixed margins $\alpha < 1$ and $\gamma < 1$ justify both absorptions. The scratch space is restored exactly at each stage. Enlarging the constant covers the finitely many remaining input lengths. $\square$

11.4 Packing tensor powers and counting circuits

Taking tensor powers makes nearby states easier to distinguish. We therefore pack one-copy states at distance of order $1/\sqrt{t}$; their tensor powers remain separated by a constant.

Lemma 11.6 (Packing tensor-power states). *For $D \geq 2$ and $t \geq 1$, there is a set $\mathcal{P}$ of pure states in $\mathbb{C}^D$ with $|\mathcal{P}| \geq (4t)^{D-1}$ such that distinct $\psi, \phi \in \mathcal{P}$ satisfy*

$$d(\psi^{\otimes t}, \phi^{\otimes t}) \geq \sqrt{1 - e^{-1/4}} > 2/10.$$

Proof. For a fixed pure state ψ and a Haar-random pure state ϕ , a trace-distance ball of radius $r \in [0, 1]$ has probability

$$\Pr[d(\psi, \phi) \leq r] = r^{2(D-1)}.$$

Indeed, unitary invariance lets us take ψ to be the first basis vector. Normalize D independent standard complex Gaussian entries to generate ϕ . Their squared moduli are independent exponentials, and the first divided by their sum has the Beta(1, $D - 1$) distribution. Its upper-tail probability at $1 - r^2$ is $(r^2)^{D-1}$.

Choose a maximal set of states separated by at least $r = 1/(2\sqrt{t})$. Compactness guarantees such a finite set, and maximality makes its radius- r balls cover the pure-state space. Their volumes give $|\mathcal{P}|r^{2(D-1)} \geq 1$. For distinct members,

$$\begin{aligned} d(\psi^{\otimes t}, \phi^{\otimes t})^2 &= 1 - |\langle \psi | \phi \rangle|^{2t} \\ &\geq 1 - \left(1 - \frac{1}{4t}\right)^t \geq 1 - e^{-1/4}. \end{aligned}$$

This proves both assertions. □

Proposition 11.7 (Quantum lower bound). *Uniformly over integers $1 \leq t \leq 2^n$ as $n \rightarrow \infty$,*

$$M(n, t) = \Omega\left(nt + \frac{2^n}{n} \log_2(t+1)\right).$$

Proof. Put $D = 2^n$ and use [Lemma 11.6](#). The triangle inequality implies that one circuit output can approximate at most one of the $|\mathcal{P}|$ tensor-power targets to error $1/10$.

Let $g \geq 1$ bound the number of gates. A circuit uses at most $2g$ scratch qubits that participate in any gate; unused zero qubits can be removed. Canonically relabel the scratch qubits by their first use, keeping the nt designated output wires fixed. A gate is specified by its type in the fixed finite set $\mathcal{G}$ and at most two wire indices in a set of size $nt + 2g$. Therefore the number of circuits of size at most g is at most

$$(g+1)(|\mathcal{G}|(nt + 2g + 2)^2)^g.$$

This upper bound includes circuits that fail to restore their scratch space, so imposing the clean-scratch condition cannot invalidate it.

If every target had a circuit of size at most g , taking logarithms and comparing with $|\mathcal{P}|$ would give

$$(D-1) \log_2(4t) \leq O(g \log_2(nt + g + 2) + \log_2(g+1)).$$

For $g \leq 2^n$ and $t \leq 2^n$, the logarithm on the right is $O(n)$, which yields the logarithmic term. If $g > 2^n$, that term already follows from $\log_2(t+1) \leq n+1$.

For the remaining nt term, take $|\psi\rangle = |1^n\rangle$. Any untouched output wire stays in zero, making the output orthogonal to $|1^{nt}\rangle$. Error below one therefore requires every output wire to participate in a gate, and a gate touches at most two wires. Thus $M(n, t) \geq nt/2$. Combining the two lower bounds proves the claim. $\square$

[Propositions 11.5](#) and [11.7](#) prove [Theorem 1.4](#). The packing argument gives a worst-case lower bound; it does not identify an explicit family of hard states.

12 Implications and open problems

Secure evaluation gives a direct application of the simultaneous size and depth bound. We then examine what lower bounds say about restricted circuit architectures and specified hard functions, before returning to deterministic construction and endpoint copy rates.

12.1 Secure batch evaluation

The simultaneous size and depth bound has a standard secure-computation corollary. Fix integers $m \geq 4$ and $0 \leq c < m/3$, and assume a synchronous network of m parties with private authenticated channels and broadcast. For every public $f : \{0, 1\}^n \rightarrow \{0, 1\}$, fixed $0 \leq \gamma < 1$, and $1 \leq t \leq 2^{\gamma n}$, there is a nonuniform protocol computing $f^{\times t}$ with perfect security against a static malicious adversary corrupting at most c parties, using $O_{\gamma, m}(2^n/n)$ communication bits and $O_{\gamma, m}(n)$ rounds. The input bits may be distributed arbitrarily among the parties, and all parties receive the output vector.

To obtain this protocol, apply the Ben-Or–Goldwasser–Wigderson framework [[BGW88](#)], with the quantitative bound of [[AAY21](#), Theorem 1.2 of the full version], to [Theorem 1.3](#). Over a fixed field $\mathbb{F}_Q$ of characteristic two with $Q > m$, AND, OR, and NOT become xy , $x + y + xy$, and $1 + x$. First mapping each supplied field element u to u^{Q-1} ensures Boolean inputs at constant cost per input for fixed m . Secure evaluation then uses $O_m(S + nt)$ communication bits and $O_m(D + 1)$ rounds for a Boolean circuit of size S and depth D , including input sharing and output reconstruction. Substitute the simultaneous bounds and $nt = o_\gamma(2^n/n)$. The correct circuit and its menus are public nonuniform protocol data; scheduling and routing are evaluated on shares.

The communication bound comes from secure circuit evaluation. Circuit lower bounds do not establish its optimality among protocols: under suitable homomorphic-encryption assumptions, computationally secure protocols can limit communication according to input and output lengths and security parameters despite large local computation [[Cou18](#), Section 1]. The perfect-security lower bounds in [[DS20](#)] concern functions with circuits linear in their input length and do not settle this larger-circuit regime.

12.2 Lower bounds and the leading coefficient

For fixed $0 < \gamma < 1$ and $t = \lfloor 2^{\gamma n} \rfloor$, the normalized worst-case size has lower limiting coefficient at least 1 by one-copy counting, while [Theorem 1.2](#) gives upper limiting coefficient $1/(1 - \gamma)$. Achieving coefficient 1 at a positive fixed exponential copy rate remains unresolved. It is also open whether

the upper coefficient $1/(1 - \gamma) + o_\gamma(1)$ can be attained at depth $O_\gamma(n)$, as the proof of [Theorem 1.3](#) allows constant losses.

[Theorem D.1](#) supplies a matching lower coefficient for a restricted resource architecture. At this batch size, suppose a circuit is partitioned into at most $2^{\mu n}$ modules, for a fixed $0 < \mu < 1$, each with polynomially many input and output ports, and an exterior of W gates. If gate-count caps $S = O(2^n/n)$ and W suffice to compute $f^{\times t}$ for every f , then

$$(1 - \gamma)S + \gamma W \geq (1 - o(1)) \frac{2^n}{n}.$$

Consequently $W = o(2^n/n)$ forces the coefficient $1/(1 - \gamma)$. The modules and their wiring may depend on f . The bound applies to separately synthesized resource banks of the kind used here, but leaves unrestricted circuits open. Its proof in [Appendix D](#) cuts the largest modules at their output ports and counts the remaining circuitry.

A conditional bound for a specified function. The counting arguments above do not identify a hard function. An inversion assumption gives a bound for the following specified Boolean family. A batch can request different bits of one inverse; collecting those bits recovers the full preimage.

Proposition 12.1 (Inverse-bit reduction). *Let $(P_m)_{m \geq 2}$ be a polynomial-time computable family of permutations of $\{0, 1\}^m$. Suppose every nonuniform Boolean circuit $I : \{0, 1\}^m \rightarrow \{0, 1\}^m$ satisfying*

$$\Pr_{y \text{ uniform in } \{0, 1\}^m} [I(y) = P_m^{-1}(y)] \geq \frac{1}{2}$$

has at least $H(m)$ gates. Define a Boolean function on $m + \lceil \log_2 m \rceil$ bits by

$$h_m(y, j) = \begin{cases} (P_m^{-1}(y))_j, & j \in [m]_0, \\ 0, & \text{otherwise.} \end{cases}$$

Then, for every integer $t \geq 1$,

$$C(h_m^{\times t}) \geq \frac{H(m)}{\lceil m/t \rceil}.$$

Proof. Let a circuit of size S compute $h_m^{\times t}$ exactly. Use $\lceil m/t \rceil$ copies of it, wiring the same y into every request slot and hardwiring the indices $0, \dots, m - 1$ across those slots. Pad any remaining slots with index zero and discard their outputs. The retained outputs give $P_m^{-1}(y)$ for every y , using at most $\lceil m/t \rceil S$ gates. The assumption gives the inequality. $\square$

For $1 \leq t \leq m$, the lower bound is at least $tH(m)/(2m)$; for $t \geq m$, it is $H(m)$. Exponential inversion hardness therefore gives an exponential circuit lower bound, but this reduction supplies no further growth once $t \geq m$. The predicate h_m itself need not be efficiently evaluable. The result neither compares $C(h_m^{\times t})$ with $tC(h_m)$ nor closes the unrestricted leading-coefficient gap.

12.3 Open problems

Can the menus be constructed deterministically? [Corollary 10.4](#) constructs circuits attaining the simultaneous size and depth bounds in time polynomial in the supplied truth-table length, with high probability of correctness on all batches. It samples fixed menus without certifying their universal property. Efficient deterministic construction of such menus, or efficient certification of a sampled menu, remains open here.

Where does the plateau end? The theorem reaches $2^{(1-\varepsilon)n}$ copies for every fixed $\varepsilon > 0$, but does not treat a rate that approaches one with n . Some degradation in that regime is forced. Suppose $n \geq 2$ and f depends on all n of its variables. Restrict a circuit for $f^{\times t}$ to the union of its output cones, and write s for the remaining gate count. Every one of the tn inputs remains. Each connected component of the underlying undirected graph contains an output, so there are at most t components. The graph therefore has at least $tn + s - t$ edges, while fan-in at most two permits at most $2s$ edges. Hence

$$C(f^{\times t}) \geq t(n - 1).$$

A bound of the form $A 2^n/n$ therefore cannot hold once $t > A 2^n/(n(n - 1))$; in particular, a uniform $O(2^n/n)$ plateau must fail for $t = \omega(2^n/n^2)$. The theorem still leaves the detailed mass-production curve throughout the near-endpoint regime $t = 2^{n-o(n)}$ unresolved. Analyzing this regime requires uniform control of the code rate and circuit overhead as the parameters vary with n .

What happens near the quantum endpoint? For $t \leq 2^n$, [Proposition 11.7](#) gives $M(n, t) = \Omega(nt + (2^n/n) \log_2(t + 1))$. [Theorem 1.4](#) matches this order for each fixed rate below one, where nt is negligible, but does not settle $t = 2^{n-o(n)}$. In particular, the $\Theta(2^n)$ cost at fixed positive rates cannot continue to $t = 2^n$, where touching all output wires already costs $\Omega(n2^n)$. Matching the two-term lower bound near this endpoint, or optimizing scratch space and T count simultaneously with total gate count, remains open here. The state theorem also does not settle the corresponding optimal tradeoff for arbitrary unitary operations.

A An explicit recursive construction

The explicitly constructible greedy scheduler gives a recursive proof of [Theorem 1.1](#). The leading-coefficient bound in [Theorem 1.2](#) uses the menu scheduler from the main text. Apply the composition bound recursively. At level k , split the input into $k + 1$ approximately equal blocks, encode one block, and invoke the preceding level on the remaining k .

For $r \geq 1$, define

$$L_r(n) = \max_{f: \{0,1\}^n \rightarrow \{0,1\}} C(f^{\times r}).$$

For each f , choose a circuit for $f^{\times r}$ of size at most $L_r(n)$. If a construction supplies fewer than r live inputs, the unused inputs of this circuit may be fixed arbitrarily and their outputs ignored. Thus a bound for $L_r(n)$ also applies to every smaller number of copies.

A.1 Grouping and composition

Partition the t requests into C fixed groups, each of size at most

$$g = \left\lceil \frac{t}{C} \right\rceil.$$

Apply [Lemma 6.2](#) independently inside each group. A code coordinate is used at most once per group and therefore at most C times overall. The inequalities $\lceil t/C \rceil \leq t/C + 1$ and $2t \leq t^2/C + C$, the latter being the arithmetic–geometric mean inequality, show that the total scheduling cost is

$$C \cdot \tilde{O}_\ell(g^2 q) = \tilde{O}_\ell\left(\frac{t^2 q}{C} + Cq\right). \quad (12)$$

The construction is valid whenever (5) holds for $g = \lceil t/C \rceil$, that is, whenever

$$\left\lceil \frac{t}{C} \right\rceil q < Dq. \quad (13)$$

Increasing C reduces scheduling cost and increases the number of suffixes each resource must handle. The recursion chooses C to balance these costs.

Bounded-demand invariant. Fix f and the local-recovery gadget above. Each $H_{y,r}$ is one fixed Boolean function on d variables. Inside each request group, disjoint recovery ensures that coordinate y occurs at most once; hence the pair (y, r) receives at most one suffix from each of the C groups. Place those suffixes in the canonical group slots and pad unused slots with a fixed dummy suffix. One circuit for $H_{y,r}^{\times C}$, of size at most $L_C(d)$, serves exactly this bounded demand. There are $q^\ell b$ such pairs, and no sharing between different pairs is assumed.

Proposition A.1 (Grouped composition). *Fix $\ell \geq 2$ and the local-recovery gadget for a split $n = p + d$. Let $C \geq 1$, set $g = \lceil t/C \rceil$, and suppose that, for every group size $1 \leq h \leq g$, a Boolean circuit of size at most $S(g, q)$ maps every multiset of h information points to enumerated pairwise disjoint affine-line recovery sets. Then every $f : \{0, 1\}^n \rightarrow \{0, 1\}$ satisfies*

$$C(f^{\times t}) \leq \underbrace{q^\ell b L_C(d)}_{\text{resource circuits}} + \underbrace{CS(g, q)}_{\text{schedule selection}} + \underbrace{\tilde{O}_\ell(tq + Cq^\ell)}_{\text{routing and decoding}}. \quad (14)$$

Proof. Apply the assembly from [Proposition 7.1](#), with one scheduler per group at total cost $CS(g, q)$. Use [Lemma 6.6](#) with $M = Cq^\ell$ keys (group, y). After scatter, fixed wiring transposes the canonical key order to (y, group) , placing all suffixes for one resource together.

The bounded-demand invariant permits evaluating $H_{y,r}^{\times C}$ in each resource position, at total cost $q^\ell b L_C(d)$. Gather and decode as in [Proposition 7.1](#). Prefix mapping, routing, and decoding together cost $\tilde{O}_\ell(tq + Cq^\ell)$. $\square$

Proposition A.2 (Composition bound). *Fix $\ell \geq 2$. Let $n = p + d$ with p sufficiently large depending on ℓ , choose $q = 2^b$ minimally so that $s_0^\ell b \geq 2^p$, and suppose [Equation \(13\)](#) holds. Then every $f : \{0, 1\}^n \rightarrow \{0, 1\}$ satisfies*

$$C(f^{\times t}) \leq \underbrace{q^\ell b L_C(d)}_{\text{recursive resource evaluation}} + \tilde{O}_\ell \left(\underbrace{\frac{t^2 q}{C}}_{\text{line scheduling}} + \underbrace{tq}_{\text{incidences and decoding}} + \underbrace{Cq^\ell}_{\text{resource slots}} \right). \quad (15)$$

The induction keeps the resource term at scale $2^n/n$ while making all three overhead terms exponentially smaller.

Proof. Put $g = \lceil t/C \rceil$. The scheduling condition lets us apply [Lemma 6.2](#) with $S(g, q) = \tilde{O}_\ell(g^2 q)$. Moreover,

$$Cg^2 q \leq C \left(\frac{t}{C} + 1 \right)^2 q = O \left(\frac{t^2 q}{C} + tq + Cq \right).$$

Apply [Proposition A.1](#); the tq term is already present, and Cq is absorbed by Cq^ℓ because $q \geq 2$ and $\ell \geq 2$. $\square$

A.2 The induction invariant

For an integer $k \geq 1$, let P_k denote the assertion that, for every fixed $0 \leq \gamma < k/(k+1)$, there is a constant $A_{k,\gamma}$ such that, for all sufficiently large n , every $f : \{0, 1\}^n \rightarrow \{0, 1\}$, and every integer $1 \leq t \leq 2^{\gamma n}$,

$$C(f^{\times t}) \leq A_{k,\gamma} \frac{2^n}{n}.$$

Every comparison below has a fixed positive linear margin in its base-two logarithm. We use without further comment that such a margin absorbs the $o(n)$ terms, every fixed polynomial factor hidden by $\tilde{O}$, and the $O(1)$ changes caused by floors and ceilings.

A.3 Two equal blocks: the base case

Lemma A.3 (Two-block base). P_1 holds.

Here $C = 1$, so no mass-production hypothesis is used inside a resource. The two equal blocks make the resource term proportional to $2^{n/2} L(n/2)$; the one-half threshold comes from the quadratic scheduler term $t^2 q$.

Proof. Fix $0 \leq \gamma < 1/2$, let $\eta = 1/2 - \gamma > 0$, and suppose $t \leq 2^{\gamma n}$. Put

$$m = \left\lfloor \frac{n}{2} \right\rfloor, \quad p = m, \quad d = n - m, \quad C = 1.$$

Thus $n = 2m + O(1)$ and all recovery sets will be globally disjoint. In block units, the request and code parameters are

$$\log_2 t \leq m - \eta n + O(1), \quad \log_2 q = \frac{m}{\ell} + o(n), \quad \log_2 D_q = \left(1 - \frac{1}{\ell}\right) m + o(n).$$

Choose a sufficiently large fixed ℓ so that

$$\frac{1}{\ell} < \eta. \tag{16}$$

Since $2m/\ell \leq n/\ell < \eta n$, this gives

$$m - \eta n + \frac{m}{\ell} < m - \frac{m}{\ell}.$$

The strict linear margin now implies $tq < D_q$ for all sufficiently large n .

The resource term in Equation (15) is, by Shannon–Lupanov synthesis,

$$q^\ell bL(d) = O_\ell \left(2^p \frac{2^d}{d} \right) = O_\gamma(2^n/n).$$

The scheduler exponent is

$$\log_2(t^2 q) \leq 2m - 2\eta n + \frac{m}{\ell} + o(n) < n,$$

where the strict inequality follows from (16). Likewise, $\log_2(tq) \leq m - \eta n + m/\ell + o(n) < n$ and $\log_2 q^\ell = m + o(n) < n$. After the suppressed polynomial factors are restored, every overhead term is $2^{(1-\Omega_\gamma(1))n}$ and is negligible compared with $2^n/n$. The same strict margins absorb all rounding. $\square$

A.4 Adding one equal block

Lemma A.4 (Equal-block step). *For every integer $k \geq 2$, P_{k-1} implies P_k .*

Proof. Fix $0 \leq \gamma < k/(k+1)$. If $\gamma < (k-1)/k$, the assertion is already P_{k-1} . We may therefore assume

$$\frac{k-1}{k} \leq \gamma < \frac{k}{k+1}.$$

Let

$$\eta = \frac{k}{k+1} - \gamma > 0, \quad m = \left\lfloor \frac{n}{k+1} \right\rfloor, \quad p = m, \quad d = n - m.$$

Write $n = (k+1)m + r$ with $0 \leq r \leq k$, and choose the number of request groups to be

$$C = \left\lceil 2^{(k-1)m - \eta n/2} \right\rceil. \tag{17}$$

This choice splits the available slack in half. Dividing the requests among C groups puts each group $\eta n/2$ below the one-block exponent m ; measured on the remaining d bits, the exponent of C is also

a fixed amount below the threshold supplied by P_{k-1} . The assumption $\gamma \geq (k-1)/k$ ensures that the exponent in (17) is positive for all sufficiently large n .

Let $\beta = (k-1)/k$. Since $d = km + r$,

$$\beta d - \log_2 C = \frac{\eta n}{2} + \frac{k-1}{k}r + O(1).$$

In particular, $C \leq 2^{(\beta-\eta/4)d}$ for all sufficiently large n . The exponent $\beta - \eta/4$ is fixed and strictly below the threshold for P_{k-1} , so the inductive hypothesis gives

$$L_C(d) = O_{k,\gamma}(2^d/d).$$

By Equation (1), the resource term is therefore

$$q^\ell b L_C(d) = O_{k,\gamma,\ell}\left(2^m \frac{2^d}{d}\right) = O_{k,\gamma,\ell}(2^n/n).$$

The remaining estimates are easiest to read in block units. Since $\log_2 t \leq \gamma n = km - \eta n + O_k(1)$, the size $g = \lceil t/C \rceil$ of a request group satisfies

$$\log_2 g \leq m - \frac{\eta n}{2} + O_k(1).$$

The code parameters satisfy

$$\log_2 q = \frac{m}{\ell} + o(n), \quad \log_2 D_q = \left(1 - \frac{1}{\ell}\right)m + o(n).$$

Choose a sufficiently large fixed ℓ so that

$$\frac{2}{(k+1)\ell} < \frac{\eta}{2}. \tag{18}$$

Then $2m/\ell < \eta n/2$, and hence

$$m - \frac{\eta n}{2} + \frac{m}{\ell} < m - \frac{m}{\ell}.$$

Together with (3), this is exactly the strict exponent margin required for Equation (13).

Finally, the complete overhead ledger in Equation (15) is

$$\begin{aligned} \log_2 \left(\frac{t^2 q}{C} \right) &\leq (k+1)m - \frac{3\eta n}{2} + \frac{m}{\ell} + o(n) < n, \\ \log_2(tq) &\leq n - m - \eta n + \frac{m}{\ell} + o(n) < n, \\ \log_2(Cq^\ell) &\leq n - m - \frac{\eta n}{2} + o(n) < n. \end{aligned}$$

The first strict inequality uses $(k+1)m \leq n$ and (18); the other two use $m = \Theta_k(n)$ and $\ell > 1$. Thus every overhead term is exponentially smaller than $2^n/n$. This proves P_k . $\square$

Alternative proof of Theorem 1.1. By Lemmas A.3 and A.4, P_k holds for every $k \geq 1$. Fix $0 \leq \gamma < 1$ and choose k so that $\gamma < k/(k+1)$. Then P_k gives the asserted bound beyond a threshold N_γ . For the finitely many integers $1 \leq n < N_\gamma$, use $C(f^{\times t}) \leq tC(f)$ and enlarge A_γ to dominate $ntC(f)/2^n$ over all relevant n, f , and t . This proves the statement for every $n \geq 1$. $\square$

B Circuit implementation details

This appendix records the gate-level details behind the scheduler and routing lemmas. The main proof uses only their statements and costs.

B.1 Projective-direction indexing

Proof of Lemma 6.1. Normalize a nonzero vector so that its first nonzero coordinate is one. The normalized directions whose first nonzero coordinate is in position $h \in \{1, \dots, \ell\}$ have the form

$$(0, \dots, 0, 1, w_{h+1}, \dots, w_\ell)$$

and form a block of size $q^{\ell-h}$. Let

$$B_h = \sum_{r=1}^{h-1} q^{\ell-r}, \quad \text{val}_q(w_{h+1}, \dots, w_\ell) = \sum_{r=h+1}^{\ell} \text{int}(w_r) q^{\ell-r},$$

where $\text{int}(w_r) \in [q]_0$ is the canonical binary representation. The rank is

$$B_h + \text{val}_q(w_{h+1}, \dots, w_\ell).$$

The blocks are consecutive and their total size is D_q .

To rank, find the first nonzero coordinate, invert it, normalize the vector, and concatenate the b -bit tail coordinates. To unrank, compare the rank with the ℓ hardwired block intervals, subtract the appropriate B_h , and split the remainder into base- q digits. Since $q = 2^b$, the last step is bit slicing. The field operations and comparisons from the circuit conventions give size $\text{poly}_\ell(b)$ in both directions. $\square$

B.2 The least-missing-direction circuit

Proof of Lemma 6.2. At step j , fewer than D_q directions are forbidden, so a valid direction exists. This proves the combinatorial statement.

For the circuit bound, represent a projective direction by normalizing its first nonzero coordinate to one and use the ranking from Lemma 6.1. At stage j , for each $y \in U$, output the sentinel rank D_q when $y = u_j$ and otherwise output the projective rank of $y - u_j$. At the gate level, first multiplex a fixed nonzero vector in place of $y - u_j$ when $y = u_j$, rank that always-nonzero vector, and then overwrite the result by D_q on the equality branch. Use $\lceil \log_2(D_q + 1) \rceil$ bits per rank. There are $(j-1)(q-1) = O(jq)$ such entries.

Sort these ranks as $r_1 \leq \dots \leq r_s$. The least missing rank can be found without a dense D_q -bit table. Rank zero is a candidate if $s = 0$ or $r_1 > 0$; after position i , rank $r_i + 1$ is a candidate if

$$r_i < D_q - 1 \quad \text{and} \quad (i = s \text{ or } r_{i+1} > r_i + 1).$$

These tests are applied to the full sorted list, including duplicates and sentinels. Only the last occurrence of a non-sentinel rank can generate its successor, and a sentinel cannot generate a candidate.

A left-to-right prefix-OR circuit identifies the first candidate in this order. Multiplexers select its rank using $O(s \log D_q)$ gates.

Since $|U \setminus \{u_j\}| < D_q$, fewer than D_q distinct non-sentinel ranks are forbidden, so a candidate exists. Sorting, comparisons, and priority selection therefore cost $\tilde{O}_\ell(jq)$ gates. Unrank the selected direction using [Lemma 6.1](#) and enumerate its $q-1$ non-target line points, at cost $q \text{poly}_\ell(b)$. Unrolling the stages and summing over $j \leq g$ gives $\tilde{O}_\ell(g^2q)$. $\square$

B.3 Four-pass fixed-wire routing

Proof of [Lemma 6.6](#). For scatter, create one slot record for each of the M table keys and combine them with the fewer than tq incidence records. Preserve a copy of every incidence record by free fan-out. The uniqueness hypothesis puts at most one incidence at each key. After the first pass in the table, a slot therefore copies the preceding suffix exactly when a type-and-key equality test succeeds; otherwise it receives a fixed dummy suffix. The predecessor of the first position is defined to be a dummy record, so the boundary case uses the same guard. The second pass places the filled slots in canonical key order, ready for the resource circuits.

For gather, combine the preserved incidences with one value record per key. The third pass places that value immediately before the unique matching incidence; a guarded local multiplexer copies it into the incidence. The fourth pass puts the first $t(q-1)$ positions in the fixed request-and-line order required by the decoders. Thus absent incidences, the first array position, and all type boundaries are handled by fixed dummy data and the same guarded update, not by data-dependent wiring.

This is a constant number of sorting networks on $O(tq + M)$ records. Each record has width $\text{poly}_\ell(n, \log q, \log(t+2), \log(M+2))$, so the standard primitives from the circuit conventions give the stated bound. $\square$

Relation to multiselection. The gather half is also an instance of multiselection. Equivalently, it is a partial-function inner join in the sense of Holmgren and Rothblum [[HR24](#), Proposition 25 and Sections 6.1–6.2]. Write M for the number of value slots and $R = t(q-1)$ for the number of incidences. Uniqueness of incidence keys gives $R \leq M$. For the resource tables used here, a key’s field coordinates and any group or code number determine its slot index by fixed arithmetic. Each incidence in canonical request–line–position order supplies this index; applying their binary main theorem independently to the b bit positions therefore gives size $O(bM + bR \log^3(M+2))$ [[HR24](#), Main theorem], within the bound of [Lemma 6.6](#). The four-pass construction handles scatter and gather within a single routing bound.

C Formalization and exact finite accounting

This appendix connects the asymptotic proof in [Section 9](#) with the Lean theorem. It records how the formal construction tracks requests, counts resource circuits, and handles arbitrary input lengths using exact integer parameters.

Companion and reproducibility. At revision [8dd82c9](#), the Lean 4.33.1/mathlib 4.33.1 module `Algebraic.MassProduction` exposes `BlockInduction.exponentialMassProduction` for the ex-

PLICIT proof and `Nonuniform.realSharpMassProduction` for the nonuniform proof. Both endpoints use only Lean’s standard axioms, with no proof placeholders. A small executable Python model of the menu verifier and the high-rate subcode, with exhaustive checks on tiny parameters, accompanies the manuscript source at revision [1c880d9](#).

Request identity and inactive slots. Each request carries a distinct fixed identifier, even when its target and suffix coincide with another request’s. The scheduler maintains a permutation of accepted and pending requests, together with disjoint recovery lines for the accepted prefix. Each phase preserves all original payloads. For regular array sizes, a line uses q scalar slots: the zero scalar is marked inactive, and its recovered Boolean contribution is fixed to zero. After XOR recovery, only the identifier and the one-bit answer need to be sorted back into request order. Thus restoring output order does not require another permutation of the full point lists.

Charge exactly the resource bank. A single hardwired table indexed by the source prefix can return the code number, information point, and basis-bit selector together. Batched lookup handles repeated prefixes and removes runtime division from this version of the construction. The resource key is (code number, point, basis bit); valid scheduled incidences have distinct keys. Evaluate precisely the $R = BNb$ Boolean resource circuits, once each. Padding the sorting networks adds inactive routing records, not resource evaluations. This distinction avoids a constant-factor loss in the main term. If $\widehat{L}(d)$ bounds the chosen one-copy synthesis circuits, the full cost is bounded by

$$R\widehat{L}(d) + H, \quad H \leq c_\ell 2^p n^7$$

in the parameter regime below, with c_ℓ independent of n, f, t . For each fixed positive integer Q , the actual synthesis circuits eventually admit $\widehat{L}(d) = \lfloor (Q+1)2^d/(Qd) \rfloor$. This checked polynomial envelope is coarser than the optimized pass count in [Theorem 6.5](#); both give the required negligible overhead.

Whole field blocks, arbitrary input lengths. One exact parameter choice fixes positive integers h, ℓ, G, S , with $\ell \leq 2^h$, and sets

$$P = \ell h + 1 < S, \quad m = \lfloor n/S \rfloor, \quad b = hm, \quad p = Pm, \quad d = n - p.$$

Choose $G + h + 1 \leq h(\ell - 1)$ and $\gamma < G/S$. For sufficiently large m , the scheduler batch 2^{Gm} covers the requested copies, including rounding, and has $512 2^{Gm} q \leq D_q$. Also $2^{Gm} q \leq 2^p$. Put the incomplete input block into the suffix; then

$$p + d = n, \quad \frac{d}{n} \geq \frac{S - P}{S} > 0.$$

Integer slopes can make P/S arbitrarily close to γ from above while keeping all these inequalities strict. All dimensions and slopes are fixed before n varies.

For $A = 2^{\ell h}$ the retained dimension on this subsequence is exactly $K = A^m - (A - 1)^m$. At a fixed positive integer precision Q , the elementary rate estimate is

$$m \geq Q(A - 1) \implies QA^m \leq (Q + 1)K.$$

The formal packing uses $B = \lfloor 2^p/(Kb) \rfloor + 1$, charging at most one extra codeword. Since $Nb = 2^{\ell h m} h m$ and $p = (\ell h + 1)m$, eventually $QNb \leq 2^p$. Consequently $QR \leq (Q + 2)2^p$. The number B

need not be polynomial in this alternative parameter choice; the bounds on the total resource bank and the key widths are what matter. Finally, $m \leq d$ and every fixed polynomial in n is eventually smaller than 2^m . These facts absorb H after normalization by $n/2^n$.

Uniform quantifiers. For a rational rate $u/v < 1$ and every positive integer precision J , the checked integer endpoint supplies one cutoff, uniform over f and all positive $t \leq 2^{\lfloor un/v \rfloor}$, with

$$J(v - u)n C(f^{\times t}) \leq (J + 1)v2^n.$$

The final real-rate theorem chooses a slightly larger rational rate and a large enough precision inside the requested additive error. Upward exponent rounding is covered by the scheduler's strict capacity margin. The result has exactly the real-rate and ε quantifiers of [Theorem 1.2](#), without substituting growing dimensions into a fixed-dimension estimate.

D A lower bound for modules with polynomial interfaces

We count the functions representable by circuits built from modules whose input and output interfaces are small. The internal gates of different modules are disjoint. Each module receives all external signals through its input ports, and its internal gates may be used outside the module only through its designated output ports. All other gates form the *exterior*. The assembled circuit is acyclic; wiring, constants, and output designations remain free.

Theorem D.1 (Module counting bound). *Fix $0 < \gamma, \mu < 1$ and polynomial bounds $a(n), b(n) \geq 1$. Put $N = 2^n$ and $t = \lfloor 2^{\gamma n} \rfloor$. Let $\mathcal{F}_n(S, W)$ be the family of functions $f : \{0, 1\}^n \rightarrow \{0, 1\}$ for which $f^{\times t}$ has a circuit with at most S gates, partitioned into at most $2^{\mu n}$ modules and an exterior of at most W gates, with at most $a(n)$ input and $b(n)$ output ports per module. For $0 \leq W \leq S = O(N/n)$,*

$$|\mathcal{F}_n(S, W)| \leq 2^{n((1-\gamma)S + \gamma W) + o(N)}.$$

The error is uniform under the fixed rate, port, and size bounds. In particular, gate-count caps covering every f must satisfy

$$(1 - \gamma)S + \gamma W \geq (1 - o(1)) \frac{N}{n}. \quad (19)$$

Any fixed positive gap below N/n on the left makes the fraction of representable functions exponentially small in N .

The proof cuts the largest modules at fewer than t output ports. For each fixed remainder, their possible replacements account for only $2^{o(N)}$ functions. The modules left behind are small enough to describe at a lower cost per gate. Two elementary counting facts make these steps precise.

Lemma D.2 (Cutting module outputs). *Fix the exterior, all interconnections, and all but a selected group of modules whose output ports total $r < t$. Let $\mathcal{F}$ be the family of functions f for which some choice of the selected modules completes the circuit to $f^{\times t}$. Then*

$$|\mathcal{F}| \leq \sum_{i=0}^r \binom{N}{i} \leq (r + 1)N^r.$$

Proof. Regard a Boolean function as a subset of its N -point domain. A set of points is *shattered* if the family realizes every Boolean labeling of those points. If $\mathcal{F}$ shattered $r + 1$ distinct points, place them in the first $r + 1$ request slots of a fixed batch and fill the remaining slots arbitrarily. Cut the selected output ports and treat their values as r independent Boolean sources. There are at most 2^r resulting output vectors from the fixed remainder, whereas shattering requires 2^{r+1} labelings of those points. This is impossible. Cutting all selected outputs also handles connections between selected modules.

Thus the VC dimension of $\mathcal{F}$, the largest size of a shattered set, is at most r . Sauer's bound [Sau72] gives the first inequality. For completeness, its induction splits a family by the last domain point, with projected families $\mathcal{F}_0, \mathcal{F}_1$. The original cardinality is $|\mathcal{F}_0 \cup \mathcal{F}_1| + |\mathcal{F}_0 \cap \mathcal{F}_1|$. The union has VC dimension at most r and the intersection at most $r - 1$, so Pascal's identity gives the binomial sum. The empty-family and zero-dimension cases supply the induction bases. Finally $\binom{N}{i} \leq N^i$ proves the second inequality. $\square$

Lemma D.3 (Descriptions with several outputs). *After deleting unused gates, circuits with A named input sources, Q designated outputs, and u gates can be described, up to relabeling of internal gates, in at most $12^u 2^Q (u + A + 2)^{u+Q}$ ways. The base-two logarithm of this count is at most*

$$u \log_2 12 + Q + (u + Q) \log_2 (u + A + 2).$$

Proof. Traverse the output cones depth first, with the roots in output order. At each root or incoming edge, record whether a new gate is discovered or an existing gate, input, or constant is named. There are at most $2u + Q$ such flags. Exactly u positions discover a gate, whose type has three possibilities, leaving at most $u + Q$ names from a set of size $u + A + 2$. Number gates in discovery order. Padding unused description positions gives the stated count; invalid descriptions can only increase it. $\square$

Proof of Theorem D.1. First fix exact gate counts S, W and a module count $M \leq 2^{\mu n}$. Pad the ports to the bounds $a = a(n)$ and $b = b(n)$, rounding these bounds up to integers if needed. Put $B = S - W$ and

$$k = \left\lfloor \frac{t - 1}{b} \right\rfloor.$$

Remove the k largest modules by gate count, or all modules if there are fewer. They expose at most $kb < t$ output ports. For each fixed remainder, Lemma D.2 bounds the logarithm of the number of possible functions by

$$\log_2(r + 1) + r \log_2 N = O(tn) = o(N).$$

This step accounts for the removed modules without describing their potentially large interiors.

Every remaining module has at most $B/(k + 1)$ gates. Applying Lemma D.3 to their a named inputs and b outputs shows that the logarithm of their total description count is at most

$$B \log_2 \left(\frac{B}{k + 1} + a + 2 \right) + O(B) + O(Mb \log(S + a + 2)).$$

Here $\log_2(k + 1) = \gamma n + O(\log(n + 2))$. Since a, b are polynomial and $\mu < 1$, the displayed bound is at most

$$(1 - \gamma)nB + o(N).$$

This remains true if no modules remain. Specifying module gate counts and the removed subset costs $O(M \log(S + 1) + M) = o(N)$ additional bits.

To include all interconnections, describe the exterior as a circuit on the nt original inputs and the Mb module output ports, with designated outputs supplying the Ma module input ports and the t final outputs. In [Lemma D.3](#), take

$$A_{\text{ext}} = nt + Mb, \quad Q_{\text{ext}} = Ma + t, \quad u = W.$$

This description includes direct wires between modules, constant connections, and output designations. Because $\mu, \gamma < 1$, the terms involving Q_{ext} are $o(N)$, while $\log_2(W + A_{\text{ext}} + 2) \leq n + O(\log(n + 2))$. The exterior therefore contributes at most $nW + o(N)$ to the logarithm. Descriptions that create cycles when the modules are reattached merely overcount the valid circuits.

For each such description, the cut bound supplies at most $2^{o(N)}$ functions. Multiplication of the counts gives

$$\log_2 |\mathcal{F}_n(S, W)| \leq (1 - \gamma)n(S - W) + nW + o(N) = n((1 - \gamma)S + \gamma W) + o(N).$$

Summing over exact counts up to the caps adds only $o(N)$ to the logarithm; the final expression is increasing in both caps. There are 2^N truth tables, proving [\(19\)](#) and the fixed-gap assertion. $\square$

When $W = o(N/n)$, the required coefficient of S is at least $1/(1 - \gamma)$. Conversely, if $S \leq (1 + o(1))N/n$ suffices for every function, [\(19\)](#) forces $W \geq (1 - o(1))N/n$ and hence $S - W = o(N/n)$. Thus coefficient one cannot be attained by keeping a positive fraction of the leading cost inside such a bank of modules. Substantial sharing across module interiors, larger interfaces, or a leading-order exterior can escape these hypotheses; the theorem makes no unrestricted lower-bound claim.

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This project began in an interactive conversation on ChatGPT.com that was exported to the author's computer as a $\text{\LaTeX}$ document. Later interactive sessions used Claude Fable 5 and 5.1 through Claude Code and ChatGPT 5.6 Sol through Codex. These systems materially influenced the mathematical experiments, broad literature searches, proof strategy, and internal constructions, including the scheduling and recursive-composition approach; their role went beyond prose editing. The author selected the directions to pursue, checked and revised the proofs and literature claims, and takes full responsibility for the correctness, originality, and integrity of the paper. Codex also assisted with development and testing of the nonuniform menu scheduler and the high-rate-code refinement, and with their Lean formalization through the sharp real-rate theorem. Codex also assisted with the quantum state preparation application and its packing lower bound.

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